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LONGRCA BENCH: ROOT-CAUSE LOCALIZATION IN LONG-HORIZON AGENT TRAJECTORIES

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ABSTRACT

In long agent executions, an early error can persist through later actions and checks, while evidence needed to trace its origin is dispersed across the history. Short histories offer limited tests of recovering error origins across substantial subsequent execution. We introduce **LongRCA Bench**: 1,140 complete failed trajectories from five sources, all human-annotated for responsible roles and earliest decisive root-cause steps. Reference roots precede completion by a median of 48 recorded steps; 28.4% of trajectories contain over 100 subsequent steps. With DeepSeek-V4-Flash on the full benchmark, the strongest of five evaluated baselines achieves 13.2% exact root-step accuracy. We propose **Root-Cause Trajectory Attribution (RCTA)**, a training-free method that organizes original candidate records and explicit handoff instructions for attribution. Segment summaries and a trajectory outline guide candidate retrieval; available handoff records supply upstream instruction context for the final instruction–execution comparison. With the same backbone and scoring protocol, RCTA reaches 24.1% exact root-step accuracy and 51.1% responsible-role accuracy. **LongRCA-Mini** provides 200 fixed trajectories for lower-cost comparative screening. Even with RCTA, fewer than one quarter of reference roots are recovered exactly.

1 INTRODUCTION

Failure attribution guides targeted debugging (Zhang et al., 2025). In long executions, evidence spans plans, tool results, and verification (Barke et al., 2026). In Figure 2, repeated checks retain a nightly lodging price as a two-night total despite correction of a separate arithmetic error. Diagnosis requires recovering dispersed evidence and separating error introduction from propagation or missed correction.

Existing benchmarks establish agent-, step-, and root-cause attribution targets (Zhang et al., 2025; Liu et al., 2026; Chen et al., 2026; Wang et al., 2026a). Short histories (Table 1) offer limited tests of recovering error origins across substantial subsequent activity, while differing recording granularity prevents treating length alone as a controlled difficulty measure.

We introduce **LongRCA Bench**[1] for retrospective attribution in 1,140 complete, human-annotated failed trajectories from five sources. The responsible workflow role and root-cause step are predicted and scored independently. The root is the earliest recorded step introducing a decisive error relevant to the failure and unrepaired at completion. Reference roots precede completion by a median of 48 recorded steps; 28.4% of trajectories contain more than 100 subsequent steps.

[1]Project page and leaderboard: `https://longrca-bench.github.io/`

As shown in Figure 1, the five evaluated baselines achieve 30.4–36.4% exact root-step accuracy on Who&When (Zhang et al., 2025) but only 2.8–13.2% on LongRCA using DeepSeek-V4-Flash (DS-V4-Flash). The datasets also differ in domains and workflows, so this gap does not isolate a trajectory-length effect.

Locating a failure-relevant action does not establish whether it introduced the error or inherited it from an earlier instruction. We propose **Root-Cause Trajectory Attribution (RCTA)**, a training-free method that first retrieves original candidate evidence from long logs, then compares it with available recorded upstream instructions to distinguish inherited errors from errors newly introduced during execution. This separates locating failure-relevant activity from identifying where the decisive error originated.

With the same DS-V4-Flash backbone and scoring protocol, RCTA raises exact root-step accuracy from the strongest evaluated baseline's 13.2% to 24.1% (Table 2). Nevertheless, fewer than one quarter of reference roots are recovered exactly.

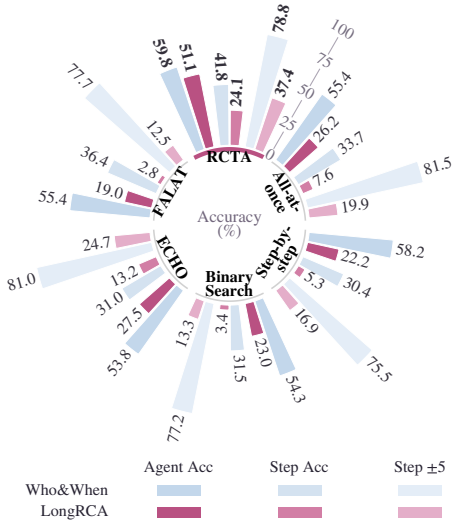


Figure 1: All six evaluated methods score lower on LongRCA across Agent Acc, Step Acc, and Step ±5. Accuracy (%) uses DS-V4-Flash on the complete Who&When ($n = 184$) and LongRCA ($n = 1{,}140$) sets, with each trajectory equally weighted within its dataset.

• **A benchmark for tracing error origins.** LongRCA Bench provides 1,140 human-annotated failed trajectories from five sources, accompanied by **LongRCA-Mini**, a fixed 200-trajectory subset for lower-cost preliminary comparisons. Headline results and leaderboard evaluation use the full benchmark.

• **A method for tracing errors to their source.** RCTA retrieves relevant original records from long logs. It then checks available earlier instructions to judge whether an error was already present in those instructions or was introduced during execution.

• **A comparison of six attribution methods.** We compare six methods on Full and across five backbones on Mini, and test RCTA with individual components removed. The results show where localization remains difficult and how individual components affect accuracy, the evidence retained, and output validity.

2 Related Work

Who&When formulates attribution as identifying a responsible agent and a decisive step (Zhang et al., 2025). AgentRx targets the earliest critical failure from which an agent does not recover, while RootSE, introduced with TrajAudit, annotates earliest decisive errors in repository-level coding traces (Barke et al., 2026; Wang et al., 2026a). Failure as a Process separately marks decisive errors, empirical unrecoverability, and first observable signs (Zhao et al., 2026). Table 1 compares these resources. HORIZON varies intrinsic task horizon under controlled conditions to study performance and failure patterns (Wang et al., 2026b). This horizon is distinct from recorded trajectory length. LongRCA Bench evaluates attribution in complete failed executions with long recorded histories, especially when substantial activity follows the reference root.

Attribution methods differ in how they organize evidence and trace error origins. ECHO combines hierarchical context with analyst consensus (Banerjee et al., 2025), while SAFARI investigates candidate faults through active trace reading and search with persistent short-term memory (Zhu et al., 2026b). FALAT uses typed dependencies among decisions, tool outputs, and messages to separate error sources from downstream carriers and guide candidate pruning and verification (Rafi et al.,

Table 1: Comparison of resources for agent failure analysis. LongRCA supports responsible-role and earliest-root-step attribution over long execution trajectories collected without error injection and fully reviewed by humans.

Benchmark	**Error Source**	**Diagnostic Targets**	**Trajectories** Total (Reviewed)	**Avg. length** (steps)
RootSE (Wang et al., 2026a)	Natural	Step[a]	102 (102)	51.7
HORIZON (Wang et al., 2026b)	Natural	Type	1995 (40)	–
TraceElephant (Chen et al., 2026)	Natural	Component, Step[b]	220 (220)	–
Failure as a Process (Zhao et al., 2026)	Natural	Steps[c], Type	1184 (1184)	42.0
Who&When (Zhang et al., 2025)	Natural	Agent, Step[a]	184 (184)	22.2
Who&When Pro (Liu et al., 2026)	Synthetic	Agent, Step[a], Type	12,326 (100)	7.5
LongRCA Bench (ours)	**Natural**	**Agent, Step[a]**	**1140 (1140)**	**156.3**

[a]Earliest decisive error, [b]Earliest point at which failure becomes inevitable, [c]Three markers: decisive error, empirical lock-in, and first observable sign of error. Annotation protocols and step granularity differ across benchmarks, so mean lengths are not directly comparable measures of difficulty; – denotes an unreported mean. Appendix J details categories and counting conventions.

2026). CHIEF combines hierarchical backtracking with local, planning-control, and data-flow attribution, screening out subsequently repaired deviations (Wang et al., 2026c).

RCTA organizes attribution around retrieved original candidate records. After candidate retrieval, it adds available explicit handoffs as upstream instruction evidence, retaining the original text and step IDs. The final comparison examines whether execution inherited an erroneous instruction or introduced a new decisive error under the task requirements. This augmentation depends on logged handoff instructions; when none are recorded, attribution proceeds from the retrieved candidate evidence (Section 4). Appendix J extends the discussion of diagnostic resources and mechanisms and details the table's comparison conventions.

3 LongRCA Bench

3.1 Task definition

We study retrospective failure attribution over completed failed executions, using the full history to distinguish error introduction from subsequent execution. This retrospective setting uses the completed record rather than a trajectory prefix for online failure detection. A trajectory $H = (h_0, \ldots, h_{T-1})$ has stable 0-based indices and recorded source roles. The responsible role is the recorded workflow role judged responsible for task failure under the annotation protocol. Methods return a role $\hat{\rho}$ and root step $\hat{r} \in \{0, \ldots, T-1\}$ in separate fields, scored independently against (ρ^*, r^*). The reference role must belong to the recorded role set but need not emit the reference root step. The role prediction space is likewise limited to recorded workflow roles. Predicted roles are read directly, never inferred from the predicted step. Each reference includes a trajectory-grounded human rationale for evidence review, which is not scored.

The reference root r^* is the earliest recorded step introducing a decisive error that is relevant to the failed execution and remains unrepaired at completion. Repaired errors and later steps that merely execute, propagate, or expose an existing error are excluded. Under a handoff, the instruction step is selected if it already contains the decisive error and the recipient carries it out without repair. A later recipient step qualifies only if it introduces the decisive error through a deviation or an error absent from the instruction. Appendix G specifies the complete boundary rules.

Figure 2 illustrates the persistence of a lodging-price misinterpretation. The human reference assigns responsibility to `stay_agent` and marks step 73 within a single output whose candidate-recording calls, steps 73–77, treat nightly prices as two-night totals. The figure follows the ultimately selected property from step 74 through subsequent planning and verification. A later correction to the food/activity budget leaves its lodging total unchanged. The case illustrates why attribution requires

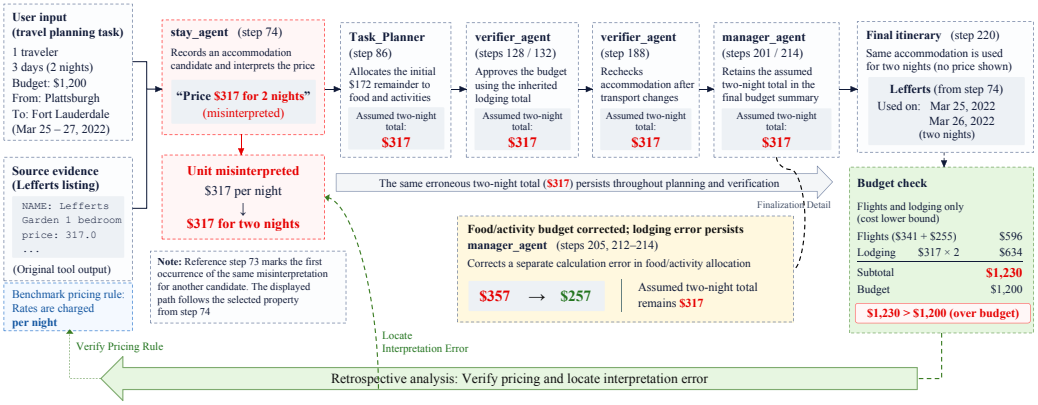


Figure 2: A price-unit misinterpretation persists through planning and verification despite a local budget correction. The human reference identifies `stay_agent` and step 73 within a single output containing candidate records at steps 73–77. The displayed path follows the ultimately selected property from step 74. Green arrows indicate retrospective diagnosis using the final itinerary and TravelPlanner's nightly pricing rule.

tracing early erroneous records through their subsequent use and distinguishing persistent errors from local repairs. Appendix K.2 details the reference annotation, the selected-property path, and the budget diagnosis.

3.2 Source Curation and Trajectory Normalization

We retain evaluator-confirmed failed task attempts from SWE-bench Pro (SWE, 128), Terminal Bench 2 (TB2, 42), TravelPlanner (TP, 685), VitaBench (Vita, 108), and WebArena Verified (WA, 177) (Deng et al., 2025; Merrill et al., 2026; Xie et al., 2024; He et al., 2026; El Hattami et al., 2025). These sources span different task domains and workflows, including fixed-role teams, group-chat coordination, and sequential execution. Failures arise from ordinary execution rather than injected errors. Infrastructure, smoke-test, and debug runs are excluded because they are not completed task attempts. Appendix E documents source configurations and trajectory generators.

Normalization preserves original content, roles, and temporal order with stable 0-based indices, without synthesizing intermediate actions or artificially subdividing source events. The shared interface does not imply identical event granularity. Methods receive only `question_ID` and `history`, including the task instruction; reference role, step, and rationale are withheld. Source evaluator outcomes guide selection and annotation, while feedback already recorded within history is retained (Appendix D).

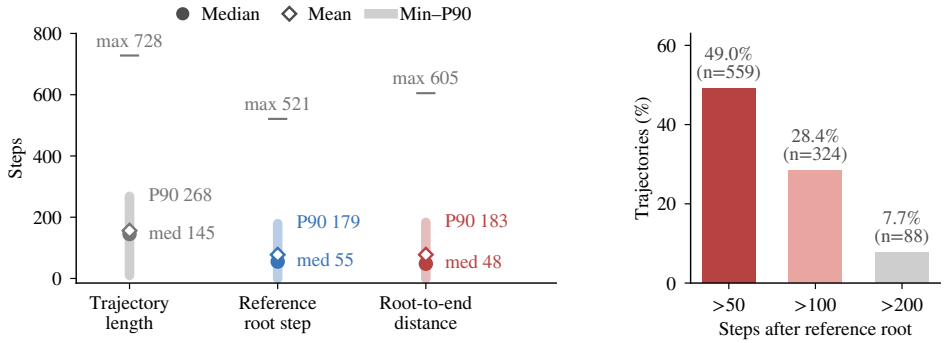
3.3 Human Annotation and Quality Control

Twenty-two master's and doctoral students in computer science annotated the trajectories. Calibration on 100 trajectories refined history navigation, root-cause search, and evidence recording, while the root definition and decision rules remained unchanged for full-scale annotation. The same annotator pool participated in both stages. Annotators independently reviewed the task, complete history, and source evaluator outcome to submit a responsible role, root step, and supporting rationale. DeepSeek V4 Flash Preview supplied factual activity summaries for navigation without root-cause judgments. Reference labels and rationales required original trajectory evidence.

Full-scale annotation provides 1,343 completed submissions with verifiable role, step, and rationale fields over 1,041 trajectories. The 294 multiply annotated trajectories yield 310 pairs: role agreement is 66.1%, step exact agreement is 40.3%, and ± 5 agreement is 50.3%; 38.1% of pairs differ by more than 20 steps. With DS-V4-Flash, RCTA ranks first in Step Acc in all 10,000 draws of existing human labels on the 294-trajectory subset (Appendix K.1).

Where multiple valid annotations were available, labels were compared field by field and exact matches retained as consensus. Two experts reviewed disagreements against original tasks, histories, evaluator outcomes, and rationales. Final quality control re-examined all retained trajectories, including calibration cases, and removed cases with unresolved ambiguity, invalid labels, or inadequate support. Of 1,200 reviewed trajectories, 60 were excluded, leaving 1,140 human-annotated failed executions. Each final record retains independent reference fields for role and step, together with a supporting rationale.

3.4 Benchmark characteristics and LongRCA-Mini



(a) Trajectory statistics

(b) Root-to-end distance

Figure 3: Long-horizon characteristics ($n = 1,140$). (a) Medians (circles), means (diamonds), minimum–P90 ranges, and maxima of trajectory length, reference root step, and root-to-end distance. (b) Fractions with more than 50, 100, or 200 recorded steps after the reference root.

Trajectory length counts recorded steps in a completed execution (mean 156.3; median 145). Root-to-end distance, $T - 1 - r^*$, counts steps after the reference root (median 48); 28.4% of trajectories have more than 100 such steps (Figure 3). These Full summaries retain natural source proportions. This distance measures subsequent activity, not error propagation or time to observable failure. Additional statistics and length-group accuracy appear in Appendix K.3 and Section 5.3, respectively.

LongRCA-Mini supports preliminary comparisons with 200 fixed trajectories, sampled uniformly without replacement using 40 per source. Sample size is calibrated using Full predictions; individual trajectories are selected without predictions or reference labels, and observed results do not trigger resampling.

Fidelity is assessed against Full Macro-5, which weights the five sources equally (Section 5.1). Across six methods and three accuracy metrics under DS-V4-Flash, the fixed subset has mean and maximum absolute score deviations of 1.665 and 3.771 pp, respectively, using cached Full predictions. These values quantify subset-induced differences under the evaluated configuration. Appendix A provides the calibration protocol and detailed fidelity results.

4 Root-Cause Trajectory Attribution

RCTA organizes original candidate records and available handoff instructions for attribution (Figure 4). Candidate retrieval and handoff augmentation determine which evidence is presented to the attribution stage; an auxiliary layer validates output fields and cited evidence.

4.1 Stage I: Failure-relevant candidate recall

Stage I builds the candidate-information package (A1). A *segment* contains consecutive recorded steps with original IDs. Task-conditioned local summaries propose failure-relevant steps, using available preceding overlap records as context. A trajectory outline aggregates subgoals and conflicts across summaries. RCTA pools and prioritizes proposed step IDs, then retrieves their original text to form the base-candidate set before handoff augmentation. A1 combines these records, local

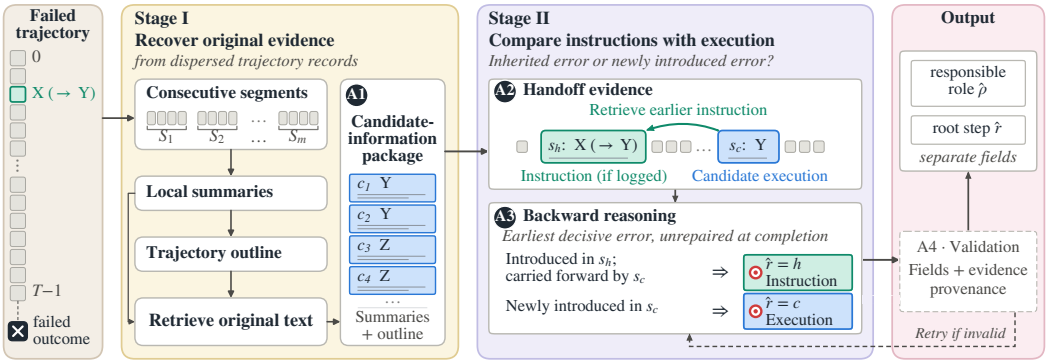


Figure 4: RCTA has two stages: Stage I constructs the candidate-information package (A1); Stage II retrieves available handoff evidence (A2) and performs backward instruction–execution reasoning (A3). Output and evidence validation (A4) checks the resulting attribution output and its evidence provenance.

summaries, and the outline, allowing attribution to inspect source evidence alongside the context guiding recall. Appendix H specifies segmentation, retrieval rules, and prompts; Algorithm 1 gives the inference sequence.

4.2 Stage II: Handoff-aware backward attribution

A *handoff instruction* is a logged message in which a sender instructs a recipient. Handoff evidence retrieval (A2) augments the base candidates with such earlier records. For executor or verifier candidates, RCTA retrieves the nearest preceding handoff whose recipient exactly matches the candidate's recorded name (the role classes and matching rules are in Appendix H). For other candidates, it retrieves the nearest earlier handoff as plan context. Candidate and handoff records retain their original text and step IDs. Instructions are attached only when present; missing instructions are not synthesized.

Backward instruction–execution reasoning (A3) evaluates candidate records against task requirements and, when available, preceding handoff instructions. It judges whether the decisive error was already present upstream or newly introduced during execution. Following Section 3.1, the decision targets the earliest supported introduction of a failure-relevant decisive error that remains unrepaired at completion. An erroneous instruction faithfully executed without repair can therefore be selected instead of its downstream candidate. Retrieved handoffs can supply root steps outside the base-candidate set. Original texts support this judgment, while summaries and the outline provide context. Without handoffs, attribution proceeds from the existing evidence.

The final attribution call returns separate responsible-role and root-step fields. Validation (A4) checks field validity and the provenance of cited handoff evidence, retaining valid fields independently. Appendix H details retries and output handling.

5 Evaluation

5.1 Experimental settings

We examine how attribution methods perform across sources and log characteristics (RQ1), how their performance varies across backbones and metrics (RQ2), and how RCTA component configurations relate to evidence retention, output validity, and localization accuracy (RQ3).

The full-benchmark comparison evaluates six training-free methods on all 1,140 failed trajectories using DeepSeek-V4-Flash (DS-V4-Flash). Alongside RCTA, All-at-once reads the complete log in one attribution prompt, Step-by-step scans it sequentially, and Binary search recursively bisects it (Zhang et al., 2025). ECHO uses hierarchical context and consensus voting (Banerjee et al., 2025); FALAT traces dependencies among decisions, tool outputs, and messages to localize error-

introducing steps (Rafi et al., 2026). All methods use the same LongRCA Bench attribution targets and scoring rules.

Agent Acc compares the explicit responsible-role output with its reference. Step Acc requires an exact match to the 0-based reference root step; Step ± 5 permits absolute step error at most five. Role and step are scored independently, with invalid fields counted as errors for the corresponding targets. Missing or unrecognized roles are invalid; missing, nonnumeric, or out-of-range steps fail both step metrics. All expected trajectories remain in accuracy denominators. Full weights trajectories equally, preserving source proportions. Mini uses $S_{\text{Macro-5}} = \frac{1}{5}\sum_{b=1}^{5} S_b$, where S_b is the current metric's source score. Both sides of Full–Mini fidelity use Macro-5. With equal source quotas, pooled accuracy equals Macro-5; this equality need not hold for MAE when valid-output counts differ.

Mini experiments share the fixed sample; fidelity reuses cached Full predictions. Ablations share segmentation and summary caches across conditions. Appendix L.1 provides normalization, MAE, and protocol details.

5.2 Full-benchmark attribution results (RQ1)

Table 2: Full-benchmark accuracies (%) using DS-V4-Flash. Overall weights trajectories equally; per-source columns report Step Acc. Bold marks the highest observed value in each column.

Method	**Overall**			**Per-source Step Acc**				
	Agent Acc	Step Acc	Step ± 5	SWE	TB2	TP	Vita	WA
All-at-once	26.2	7.6	19.9	1.6	7.1	4.5	19.4	16.9
Step-by-step	22.2	5.3	16.9	0.8	4.8	3.5	12.0	11.3
Binary search	23.0	3.4	13.3	0.8	4.8	2.0	12.0	5.1
ECHO	27.5	13.2	24.7	7.8	21.4	8.9	22.2	26.0
FALAT	19.0	2.8	12.5	2.3	2.4	2.9	3.7	2.3
RCTA	**51.1**	**24.1**	**37.4**	**38.3**	**26.2**	**18.8**	**25.0**	**33.3**

On Full with DS-V4-Flash, RCTA has the highest observed values on all three accuracy metrics, but exact root recovery remains below one quarter (Table 2). Role accuracy also requires comparison with source-specific priors: always predicting each source's most frequent reference role yields 40.3% Agent Acc, 10.9 percentage points below RCTA (Table 24). This baseline uses role frequencies from the evaluated set. Identifying the responsible role and locating the originating error remain distinct targets; success on the former does not establish success on the latter.

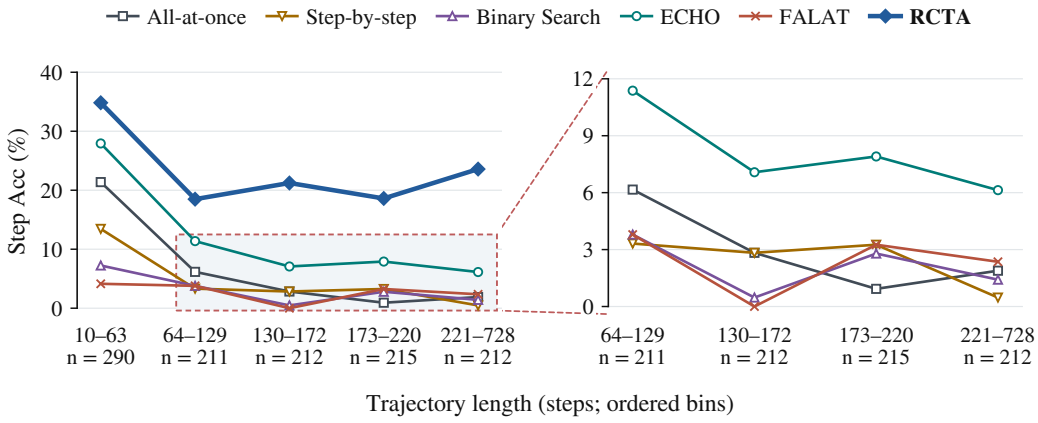
RCTA's Step Acc lead appears in every source, so its overall advantage is not confined to TravelPlanner, the largest subset. The between-source comparisons remain descriptive because the represented tasks and workflows differ. Appendix L.2 retains the complete overall metrics, including Root MAE; Appendix F supplies the source visualization and stratified numeric results.

The cross-dataset comparison in Figure 1 uses the same DS-V4-Flash backbone; Appendix C provides the protocol, results, and metric interpretation.

5.3 Localization across log characteristics (RQ1)

All methods share groups defined only by recorded length, without splitting equal lengths (Figure 5). RCTA has the highest observed Step Acc in every group, yet all six methods score lower in each longer group than in the shortest group. Variation among the longer groups is non-monotonic: RCTA rises from 18.6% to 23.6% between the final two groups.

The longest group mainly contains 221–400-step trajectories (170 of 212), so its score does not characterize only the extreme longest runs. Source and workflow composition also differ across groups, preventing interpretation as an independent causal effect of length. Appendix F gives the numeric results, partition algorithm, and four- and six-group sensitivity checks.



(a) Overall accuracy (0–40%) (b) Zoom: bins 2–5 (0–12%)

Figure 5: Step Acc on all 1,140 trajectories with DS-V4-Flash. Methods share length-only groups without splitting equal lengths. Panel (a) shows all groups; panel (b) magnifies the 0–12% region of groups 2–5. Unequal-width groups are equally spaced; connecting lines are visual guides.

Within-source analyses further examine root-to-end distance and log text volume. On TravelPlanner and VitaBench, all six methods have lower Step Acc and Step ± 5 in the highest-character group than in the lowest-character group. Patterns vary across sources, and localization performance does not consistently decline with root-to-end distance (Appendix F.1).

5.4 Cross-backbone comparison on LongRCA-Mini (RQ2)

Table 3: Agent and Step Acc (%, Macro-5) on the fixed 200-trajectory Mini (40 per source). V4 denotes DeepSeek-V4. Bold marks the highest point estimate per backbone and metric; complete metrics and paired analyses are in Appendix B.

Method	**V4-Flash**		**V4-Pro**		**GPT-5.6 Sol**		**Kimi K3**		**Qwen3.5 35B-A3B**	
	Agent	Step	Agent	Step	Agent	Step	Agent	Step	Agent	Step
All-at-once	42.5	7.5	50.0	11.5	53.0	17.0	54.5	11.0	**51.0**	3.5
Step-by-step	38.5	6.5	47.5	10.5	49.0	13.0	44.0	9.5	46.5	4.5
Binary search	40.5	7.5	45.5	10.0	51.5	15.5	42.0	12.0	40.0	5.0
ECHO	45.0	13.0	53.5	18.0	55.5	18.5	55.0	20.5	48.5	10.0
FALAT	35.5	2.5	48.0	14.5	52.0	14.0	52.0	18.0	41.0	11.0
RCTA	**60.5**	**27.0**	**61.0**	**28.0**	**64.5**	**30.0**	**61.5**	**31.5**	48.5	**13.5**

On LongRCA-Mini, the six-method comparison spans five backbones (Table 3). RCTA leads Agent Acc on four. Qwen is the exception: All-at-once leads Agent Acc and ECHO leads Step ± 5.

The comparisons do not establish significant superiority on every backbone or isolate effects of model scale. Appendix B retains all metrics, paired tests, and their endpoint-family sensitivity. Reference-label overlap also shows complementary exact localizations between RCTA and the five-baseline union on Mini (Appendix B). Across the five evaluated backbones, RCTA has the highest Step Acc point estimate, but does not lead every accuracy metric.

5.5 Component analysis (RQ3)

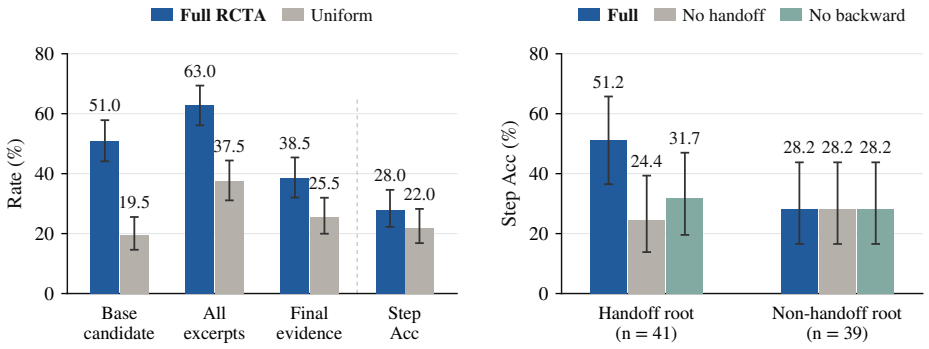
This study uses dedicated ablation inference settings, distinct from the Full-benchmark configuration (Appendix H). All conditions share trajectories, reference labels, segmentation, and summary caches. The A1 control uses equally many uniformly positioned excerpts and masks location and provenance cues, testing the complete candidate-information package. Removing A2 omits handoff evidence; removing A3 retains the original excerpts but removes the backward-attribution package;

Table 4: Paired ablations on the fixed Mini sample (Macro-5 percentages). Cov. measures valid-output coverage; p_{Holm} adjusts the four Step Acc tests.

Setting	Agent Acc↑	Step Acc↑	Step ±5 ↑	Cov.↑	p_{Holm}
RCTA (all components)	56.0	28.0	41.0	97.0	–
w/o candidate-information package (A1)	53.5	22.0	39.0	91.5	.17
w/o handoff evidence (A2)	53.5	24.5	38.0	97.0	.75
w/o backward attribution (A3)	51.0	25.0	41.0	97.5	.75
w/o validation (A4)	54.0	26.5	39.5	89.0	.75

removing A4 omits validation-driven retry and output sanitization. Appendices H and L.1 specify these controls and scoring rules.

All four removals lower Step Acc point estimates, but none passes Holm correction across the four comparisons ($p_{\text{Holm}} \geq .17$; Table 4). Non-significance does not establish equivalence; Appendix L.3 reports the paired intervals and complete results.



(a) Evidence coverage and localization

(b) Step Acc by annotated root type

Figure 6: Component diagnostics. (a) Reference-root hit rates and final Step Acc (all $n = 200$). (b) Step Acc on handoff-containing trajectories by reference root: handoff ($n = 41$) or non-handoff ($n = 39$). Full and A3 share the original excerpts. Error bars: 95% Wilson intervals; subgroup contrasts are exploratory.

The candidate-information package shows a larger gain in reference-root retention than in final Step Acc (Figure 6(a)); having the root in the evidence does not ensure correct localization. Within the 80 component-study trajectories containing explicit handoffs, the observed Step Acc differences for A2 and A3 are confined to the handoff-root subgroup (panel (b)). These exploratory comparisons cover the shared diagnosis–execution–verification workflow of SWE-bench Pro and Terminal-Bench 2; they do not evaluate trajectories without logged handoffs. Validation's clearest observed change is valid-output coverage, from 89% to 97%, with exact-root correctness changing on three of 200 trajectories.

6 Discussion and Limitations

Our evaluation addresses retrospective attribution from complete failed trajectories. A reference root need not be recognizable when it occurs; online detection requires evaluation using only the observed execution prefix, including warning timeliness and false-alarm rates. Whether warnings or interventions improve task outcomes remains untested. Evidence validation checks output fields and cited provenance, without establishing causal correctness. Agreement with human role and step labels likewise does not evaluate rationale quality or complete causal-chain reconstruction. Generalization is limited to the represented task domains, workflows, and inference settings. The comparisons do not establish an advantage under matched token or cost budgets (Appendix H).

7 CONCLUSION

LongRCA Bench provides 1,140 human-annotated long failure trajectories for independent role attribution and earliest decisive root-step localization. RCTA combines original candidate records with explicit handoff evidence for instruction–execution comparison. On the full benchmark with DS-V4-Flash, it reaches 24.1% exact root-step accuracy, compared with 13.2% for the strongest evaluated baseline. Despite this improvement, fewer than one quarter of reference roots are recovered exactly, leaving precise localization a substantial challenge.

AI USE STATEMENT

We used GPT-5.6 Sol and Codex for literature discovery, drafting, editing, and LaTeX formatting, and DeepSeek V4 Flash Preview for annotation-navigation summaries. We reviewed the AI-assisted content and references and take responsibility for the final work.

ETHICS STATEMENT

Annotators were informed about the study and participated as part of the research group's work. We screened the released data for sensitive information and reviewed the source licenses.

REPRODUCIBILITY STATEMENT

Sections 3–5 describe benchmark construction, RCTA, and evaluation. Appendices A and K.1 document the sampling manifest and annotation protocol. Prompt and inference details are provided in Appendix H, and scoring rules in Appendix L.1.

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A LongRCA-Mini sampling and reproducibility

Sampling protocol. The sample-size calibration uses master seed 20260820 and 10,000 draws for each candidate size under equal-quota simple-random sampling. The released 200-QID manifest is generated with the independent master seed 20260821. It records a SHA-256 hash of the source dataset and a hash for every selected record. Final manifest selection is without replacement and uses only source membership, QIDs, and the fixed seed; no prediction outcomes or reference labels influence which QIDs are selected. The manifest is not rerolled to improve an observed result and contains 40 QIDs per source. Full-benchmark predictions from six evaluated methods under DS-V4-Flash are used to calibrate sample size, not to select individual trajectories. All methods share the sampled QIDs within each calibration draw and the same released manifest for the fixed-subset check. Agent Acc follows the independent explicit-output scoring protocol in Section 5.

The candidate sizes are 75, 100, 125, 150, 175, 200, 225, 250, 300, 400, and 600. Equal allocation caps each source at its population size and redistributes unused slots evenly, with deterministic rounding to preserve the requested total. All sampled scores retain equal source weights in Macro-5 even when source-size caps yield unequal quotas.

Score errors and calibration criteria. Calibration and fixed-manifest fidelity use the same full-benchmark predictions. For method m and metric k, absolute score error in percentage points (pp) is

$$D_{m,k} = 100 \left| S^{\text{Mini}}_{m,k} - S^{\text{Full}}_{m,k} \right|.$$

Both scores are fractions aggregated with Macro-5. Full reaggregates all 1,140 predictions per method using equal source weights, which differ from the original-proportion weighting in Table 2. Across the 18 method–metric pairs formed by the six methods and Agent Acc, Step Acc, and Step ± 5, a candidate size passes when

$$\max_{m,k} \text{Median}(D_{m,k}) \leq 2 \text{ pp}, \qquad \max_{m,k} Q_{0.95}(D_{m,k}) \leq 6 \text{ pp},$$

where each median and quantile is computed over repeated draws for one method–metric pair before taking the maximum across pairs. These predefined criteria depend only on absolute score errors.

As shown in Figure 7 and Table 5, $n = 200$ is the smallest tested size satisfying both error criteria; $n = 175$ exceeds the median-error threshold. The empirical p95 uses the nearest-rank definition; the maximum of per-pair p95 errors is not a simultaneous 18-pair coverage bound.

Equal allocation follows the equal-source Macro-5 objective; calibration determines the smallest passing size among the tested values for that protocol, not an optimum over all possible sampling

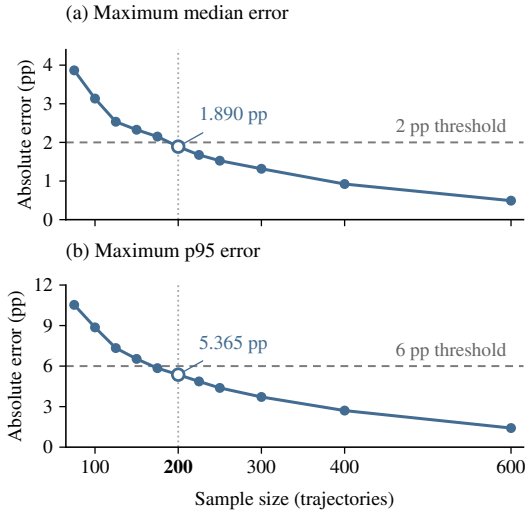


Figure 7: Mini calibration with cached predictions and 10,000 source-stratified draws per size: maximum per-pair median and 95th-percentile absolute Macro-5 errors across 18 method–metric pairs.

Table 5: Six-method sample-size calibration with equal-quota simple-random sampling. Quotas follow the source order SWE-bench Pro, Terminal-Bench 2, TravelPlanner, VitaBench, and WebArena Verified. For each of 18 method–metric pairs, absolute score errors are summarized over 10,000 draws. "Max. median" and "Max. p95" report the maximum per-pair median and 95th-percentile errors (pp). "Pass" requires only max. median ≤ 2 pp and max. p95 ≤ 6 pp.

Size	Quotas	Max. median	Max. p95	Pass
75	15/15/15/15/15	3.865	10.532	No
100	20/20/20/20/20	3.135	8.865	No
125	25/25/25/25/25	2.535	7.335	No
150	30/30/30/30/30	2.327	6.532	No
175	35/35/35/35/35	2.151	5.849	No
200	**40/40/40/40/40**	**1.890**	**5.365**	**Yes**
225	46/42/46/46/45	1.676	4.865	Yes
250	52/42/52/52/52	1.526	4.384	Yes
300	65/42/65/64/64	1.317	3.716	Yes
400	90/42/90/89/89	0.922	2.708	Yes
600	128/42/161/108/161	0.491	1.415	Yes

designs or integer sizes. A supplementary repeat with seed 20260907 and 10,000 draws per candidate size also yields $n = 175$ failing and $n = 200$ as the smallest passing tested size.

Fixed-manifest fidelity. The fixed-manifest check compares predictions on the released 200 QIDs with Full under Macro-5. Selection follows the prediction-independent rule in Section 3.4.

Table 6 reports a mean absolute deviation of 1.665 pp and a maximum of 3.771 pp across the 18 method–metric pairs. These deviations are computed before rounding the displayed scores and apply to this fixed manifest; calibration quantiles instead summarize repeated samples. The comparison measures subset-selection discrepancy under shared predictions and reference labels, not run-to-run inference variance. It supports preliminary comparisons, ablations, and configuration screening under the evaluated protocol, without establishing fidelity for future methods. Close comparisons, per-source claims, headline results, leaderboard submissions, changed data or scoring protocols, and materially different method behavior require confirmation on all 1,140 trajectories.

As shown in Table 7, the fixed Mini retains 20.11% of history characters and 18.14% of recorded steps. The Macro-5 mean trajectory length differs from Full by 6.1 steps, and the mean root-to-end distance differs by 0.5 steps. Score fidelity is summarized in Table 6.

As shown in Table 8, the maximum within-source KS distance ranges from 0.040 for Terminal Bench 2 to 0.177 for SWE-bench Pro. The largest change in mean trajectory length occurs in SWE-bench Pro, from 287.0 to 263.0 steps.

As shown in Table 9, RCTA Step Acc changes from 38.3% to 32.5% on SWE-bench Pro and from 25.0% to 32.5% on VitaBench. These source-level shifts describe variation within the fixed manifest; each Mini source contains only 40 trajectories.

For direct comparison with the full benchmark's natural source mixture, the Mini source scores can instead be post-stratified as $S_{\text{original}} = \sum_{b=1}^{5}(N_b/1140)S_b$.

In Table 10, Full and Mini use the original source proportions as their aggregation weights. This secondary estimand is separate from Macro-5 in Table 6; the two types of aggregate should not be compared as if they used the same source weights.

Reproduction artifacts. The release includes scripts/sample_equal_benchmark_dataset.py, scripts/analyze_longrca_six_method_calibration.py, scripts/analyze_longrca_mini_fidelity.py, and scripts/plot_longrca_six_method_calibration.py, together with the fixed QID manifest, repeated-draw indices and scores, and per-size and per-method–metric results. Input hashes and the Python, NumPy, and Matplotlib versions are recorded in the analysis manifests.

The calibration code verifies identical 1,140-QID coverage across the six methods, exact quotas and no within-draw duplicate QIDs, and identical draw indices and scores under same-seed replay. The sampling function accepts only source population sizes, sample size, repetition count, and seed; prediction scores are joined after drawing. The fixed-manifest fidelity check reuses the released 200 QIDs without modifying their selection.

Table 6: Fidelity of the fixed LongRCA-Mini manifest. Each cell reports **Full** $\rightarrow$ **Mini** ($|\Delta|$): Macro-5 accuracy (%) and absolute difference (pp). Across the 18 method–metric comparisons, the mean absolute deviation is 1.665 pp and the maximum deviation is 3.771 pp. All results use DS-V4-Flash.

Method	**Agent Acc**	**Step Acc**	**Step** ± 5
All-at-once	$48.2 \rightarrow 48.5$ (0.3)	$9.9 \rightarrow 8.5$ (1.4)	$23.7 \rightarrow 21.5$ (2.2)
Step-by-step	$41.8 \rightarrow 41.5$ (0.3)	$6.5 \rightarrow 6.5$ (0.0)	$19.8 \rightarrow 17.5$ (2.3)
Binary search	$40.8 \rightarrow 39.5$ (1.3)	$4.9 \rightarrow 7.5$ (2.6)	$18.1 \rightarrow 20.0$ (1.9)
ECHO	$49.4 \rightarrow 46.5$ (2.9)	$17.3 \rightarrow 13.5$ (3.8)	$31.2 \rightarrow 29.0$ (2.2)
FALAT	$33.1 \rightarrow 34.5$ (1.4)	$2.7 \rightarrow 3.0$ (0.3)	$15.5 \rightarrow 18.0$ (2.5)
RCTA	$\mathbf{60.9} \rightarrow \mathbf{61.0}$ (0.1)	$\mathbf{28.3} \rightarrow \mathbf{27.0}$ (1.3)	$\mathbf{44.1} \rightarrow \mathbf{41.0}$ (3.1)

Table 7: Structural comparison of LongRCA Bench and the fixed LongRCA-Mini manifest. The KS statistic is the maximum across within-source comparisons of trajectory length, root-to-end distance, and normalized root position. History characters are used as a model-independent input-cost proxy.

Structural quantity	**Full**	**Mini**	**Comparison**
Trajectories	1,140	200	17.54% retained
Recorded steps	178,137	32,317	18.14% retained
History characters	207,731,218	41,784,984	20.11% retained
Macro-5 mean trajectory length	167.7	161.6	-6.1 steps
Macro-5 mean root-to-end distance	104.9	104.4	-0.5 steps
Macro-5 mean normalized root position	0.420	0.399	-0.021
Maximum within-source KS distance	–	–	0.177

Table 8: Within-source structural audit of the fixed manifest. Entries are Full/Mini means; input length is measured in thousands of history characters, and normalized root position is $r^*/(T-1)$. KS is the largest empirical CDF distance across the four reported characteristics. SWE, TB2, TP, Vita, and WA denote the five sources in the order used in Table 5.

Source	**Steps**	**Input kchars**	**Root-to-end**	**Root position**	**KS**
SWE	287.0/263.0	450.2/432.9	215.4/218.5	0.250/0.181	0.177
TB2	272.1/265.6	297.2/294.9	186.9/179.5	0.263/0.268	0.040
TP	165.9/167.8	163.3/154.1	65.9/67.8	0.597/0.569	0.134
Vita	54.6/50.8	23.2/26.9	29.7/26.0	0.445/0.473	0.134
WA	58.8/60.8	131.2/135.8	26.8/30.1	0.543/0.502	0.147

Table 9: Per-source Full/Mini accuracy (%) for the fixed manifest. Each cell reports the full source score followed by the Mini source score. Source abbreviations follow Table 8. These values are an audit of source-level variation, not evidence that 40 trajectories support definitive per-source comparisons.

Method	**Metric**	**SWE**	**TB2**	**TP**	**Vita**	**WA**
All-at-once	Agent Acc	47.7/42.5	61.9/62.5	7.2/7.5	82.4/85.0	41.8/45.0
	Step Acc	1.6/2.5	7.1/5.0	4.5/5.0	19.4/20.0	16.9/10.0
	Step ± 5	4.7/7.5	21.4/20.0	15.2/10.0	41.7/42.5	35.6/27.5
Step-by-step	Agent Acc	38.3/30.0	40.5/40.0	2.8/5.0	83.3/85.0	44.1/47.5
	Step Acc	0.8/2.5	4.8/5.0	3.5/0.0	12.0/12.5	11.3/12.5
	Step ± 5	10.2/7.5	16.7/15.0	13.0/10.0	30.6/27.5	28.8/27.5
Binary search	Agent Acc	39.1/27.5	45.2/45.0	7.4/7.5	82.4/85.0	29.9/32.5
	Step Acc	0.8/2.5	4.8/2.5	2.0/7.5	12.0/17.5	5.1/7.5
	Step ± 5	3.9/5.0	23.8/22.5	9.5/15.0	32.4/35.0	20.9/22.5
ECHO	Agent Acc	45.3/32.5	59.5/60.0	7.2/5.0	82.4/85.0	52.5/50.0
	Step Acc	7.8/7.5	21.4/22.5	8.9/7.5	22.2/20.0	26.0/10.0
	Step ± 5	18.0/17.5	38.1/37.5	18.8/17.5	42.6/45.0	38.4/27.5
FALAT	Agent Acc	32.0/37.5	28.6/30.0	6.6/10.0	79.6/80.0	18.6/15.0
	Step Acc	2.3/5.0	2.4/2.5	2.9/2.5	3.7/2.5	2.3/2.5
	Step ± 5	10.9/10.0	16.7/17.5	10.1/10.0	26.9/32.5	13.0/20.0
RCTA	Agent Acc	63.3/52.5	57.1/55.0	41.2/47.5	82.4/85.0	60.5/65.0
	Step Acc	38.3/32.5	26.2/25.0	18.8/22.5	25.0/32.5	33.3/22.5
	Step ± 5	46.9/40.0	45.2/42.5	29.9/27.5	47.2/52.5	51.4/42.5

Table 10: Original-mixture fidelity audit. Cells report Full $\rightarrow$ post-stratified Mini ($|\Delta|$), in accuracy percentages and pp. Both sides use the original source weights, not Macro-5.

Method	**Agent Acc**	**Step Acc**	**Step ± 5**
All-at-once	$26.2 \rightarrow 26.6\ (0.4)$	$7.6 \rightarrow 6.9\ (0.7)$	$19.9 \rightarrow 15.9\ (4.0)$
Step-by-step	$22.2 \rightarrow 23.3\ (1.1)$	$5.3 \rightarrow 3.6\ (1.7)$	$16.9 \rightarrow 14.3\ (2.7)$
Binary search	$23.0 \rightarrow 22.4\ (0.6)$	$3.4 \rightarrow 7.7\ (4.3)$	$13.3 \rightarrow 17.2\ (3.9)$
ECHO	$27.5 \rightarrow 24.7\ (2.9)$	$13.2 \rightarrow 9.6\ (3.5)$	$24.7 \rightarrow 22.4\ (2.3)$
FALAT	$19.0 \rightarrow 21.2\ (2.2)$	$2.8 \rightarrow 2.8\ (0.0)$	$12.5 \rightarrow 14.0\ (1.5)$
RCTA	$51.1 \rightarrow 54.6\ (3.5)$	$24.1 \rightarrow 24.7\ (0.5)$	$37.4 \rightarrow 34.2\ (3.2)$

B Complete cross-backbone results on LongRCA-Mini

In the complete results in Table 11, each source contributes 40 trajectories, so aggregate accuracies equal Macro-5. Joint Exact and Joint ± 5 require both correct role attribution and the corresponding step criterion; Step MAE summarizes valid-output absolute root-step error. The entries in Table 11 are descriptive point estimates rather than repeated-run averages; paired uncertainty analyses and significance tests are reported separately below.

Table 11: Complete six-metric record for the fixed 200-trajectory cross-backbone experiment. Accuracies are Macro-5 percentages; Step MAE is the reported valid-output mean absolute root-step error in steps. Bold marks the best value within each backbone and metric (including ties). Shading identifies RCTA. Evaluation settings and prediction coverage are detailed in the accompanying text.

Method	Agent Acc ↑	Step Acc ↑	Step ± 5 ↑	Joint Exact ↑	Joint ± 5 ↑	MAE ↓
DS-V4-Flash						
All-at-once	42.5	7.5	20.5	7.5	13.0	44.915
Step-by-step	38.5	6.5	18.0	6.5	10.0	41.340
Binary search	40.5	7.5	20.5	7.5	12.0	43.465
ECHO	45.0	13.0	28.0	13.0	18.0	40.930
FALAT	35.5	2.5	17.5	2.5	10.5	53.685
RCTA	**60.5**	**27.0**	**41.0**	**27.0**	**34.5**	**28.360**
DS-V4-Pro						
All-at-once	50.0	11.5	29.5	11.5	19.5	36.200
Step-by-step	47.5	10.5	23.0	10.5	16.0	41.865
Binary search	45.5	10.0	28.5	10.0	18.0	34.555
ECHO	53.5	18.0	35.5	18.0	25.5	36.540
FALAT	48.0	14.5	23.5	14.5	20.5	52.370
RCTA	**61.0**	**28.0**	**43.0**	**28.0**	**37.0**	**31.365**
GPT-5.6-sol						
All-at-once	53.0	17.0	34.0	17.0	26.0	42.690
Step-by-step	49.0	13.0	26.0	13.0	21.0	40.635
Binary search	51.5	15.5	31.5	15.5	25.5	32.710
ECHO	55.5	18.5	32.0	18.5	24.5	44.305
FALAT	52.0	14.0	29.0	14.0	21.5	39.820
RCTA	**64.5**	**30.0**	**42.0**	**29.5**	**38.0**	**31.685**
Kimi K3						
All-at-once	54.5	11.0	29.5	10.5	20.5	34.225
Step-by-step	44.0	9.5	23.5	9.5	16.0	48.855
Binary search	42.0	12.0	28.0	12.0	20.5	40.145
ECHO	55.0	20.5	37.0	20.5	28.0	**30.760**
FALAT	52.0	18.0	30.0	18.0	24.0	41.265
RCTA	**61.5**	**31.5**	**41.5**	**31.0**	**37.5**	31.995
Qwen3.5-35B						
All-at-once	**51.0**	3.5	18.0	2.5	11.0	**42.550**
Step-by-step	46.5	4.5	18.0	3.0	12.0	44.865
Binary search	40.0	5.0	18.0	4.0	12.5	43.560
ECHO	48.5	10.0	**26.5**	9.0	19.5	44.650
FALAT	41.0	11.0	20.0	11.0	15.5	51.345
RCTA	48.5	**13.5**	25.0	**13.5**	**21.0**	45.330

Evaluation protocol and prediction coverage. DS-V4-Flash, DS-V4-Pro, GPT-5.6 Sol, and Kimi K3 each have 200 unique predictions per method on the shared 40-per-source sample, providing 4,800 method–trajectory records for the paired analysis below. Kimi K3 is supported by complete per-trajectory predictions, with the same fixed accuracy denominators as the other backbones. The table retains the MAE exception on Kimi K3: ECHO obtains 30.760 steps, compared with 31.995 for RCTA. The paired RCTA–ECHO MAE difference is 1.235 steps (95% CI: $[-6.975, 9.235]$), so the point-estimate reversal is not a resolved MAE difference. Exact-localization gains need not imply superiority on every reported measure.

Qwen3.5-35B-A3B has 1,200 valid method–trajectory records (200 per method, 40 per source), with no missing, duplicate, or parse-failed predictions. Its scores use explicit role and root-step outputs; RCTA reads `root_cause.agent/step`. On this backbone, RCTA also leads Joint Exact (13.5%) and Joint ± 5 (21.0%). Although ECHO has higher Step ± 5 (26.5% versus 25.0%), RCTA has higher Joint ± 5 (21.0% versus 19.5%), indicating more cases where role and tolerant step predictions are simultaneously correct. This aggregate observation alone does not establish a mechanism of improved role–step consistency.

Paired Step Acc comparisons. All paired comparisons retain the same 200 trajectories, with invalid predictions counted as errors. Table 12 reports pointwise 95% source-stratified paired-bootstrap intervals from 20,000 resamples and two-sided exact McNemar tests. Holm-20 adjusts the 20 Step Acc comparisons of RCTA with five baselines on four backbones; Holm-40 adjusts the combined 40-comparison family of Agent Acc and Step Acc. Both are reported to show sensitivity to the endpoint family, rather than treating pointwise intervals as simultaneous guarantees. These analyses use one retained prediction per method and backbone; resampling trajectories does not capture inference-time variability. Qwen3.5-35B-A3B is not included in this paired analysis.

Table 12: Paired RCTA–ECHO Step Acc differences on LongRCA-Mini ($n = 200$). Δ and pointwise 95% intervals are in percentage points; W/L counts RCTA-only/ECHO-only correct predictions. Holm-20 adjusts the Step Acc family; Holm-40 jointly adjusts Agent Acc and Step Acc.

Backbone	Δ	95% CI	W/L	Holm-20 p	Holm-40 p
DS-V4-Flash	$+14.0$	$[7.5, 20.5]$	39/11	0.00072	0.00217
DS-V4-Pro	$+10.0$	$[3.5, 17.0]$	36/16	0.00779	0.06007
GPT-5.6 Sol	$+11.5$	$[4.5, 19.0]$	40/17	0.00640	0.04162
Kimi K3	$+11.0$	$[5.0, 17.0]$	31/9	0.00322	0.01155

For DS-V4-Pro, the adjusted Step Acc comparison changes from $p = .00779$ under Holm-20 to $p = .060$ under Holm-40; the other three RCTA–ECHO comparisons remain below .05 under both families. A source-stratified task-cluster bootstrap that keeps repeated attempts together also gives positive pointwise RCTA–ECHO Step Acc intervals on all four backbones (184 task groups among 200 trajectories). These checks concern the evaluated samples and settings, not unseen benchmarks or backbones.

Sensitivity to source composition. For Table 13, we average the four model outcomes within each trajectory and then average over the retained benchmarks. Excluding each benchmark in turn leaves a mean gain of 8.3–13.0 percentage points on the remaining Mini trajectories. This is a sensitivity of the observed aggregate to source composition. Bootstrap resampling keeps a trajectory's model outcomes together and preserves the retained source proportions; the five intervals are pointwise, not simultaneous coverage guarantees.

Complementary correct localizations. Figure 8 compares RCTA with the union of five baseline correct sets separately for each backbone on LongRCA-Mini. Both directions of complementarity are retained: RCTA correctly localizes 23–35 trajectories per backbone that all five baselines miss, while the baselines correctly localize 27–40 that RCTA misses. RCTA's total correct count combines the "Both" and "RCTA only" segments; the latter measures unique coverage relative to the baselines, not total accuracy. The union is a gold-informed set summary, not the performance of a deployed ensemble or a cost-matched competitor. The four rows reuse the same 200 Mini trajectories and must not be added to count unique instances.

Table 13: Leave-one-benchmark-out sensitivity of the four-backbone mean RCTA–ECHO Step Acc gain. Each row retains 160 trajectories from four benchmarks; differences and intervals are in percentage points.

Excluded benchmark	Mean gain	95% CI
SWE-bench Pro	+8.3	[3.6, 13.1]
Terminal-Bench 2	+13.0	[8.6, 17.5]
TravelPlanner	+11.4	[6.4, 16.4]
VitaBench	+13.0	[7.8, 18.1]
WebArena Verified	+12.5	[7.3, 17.8]

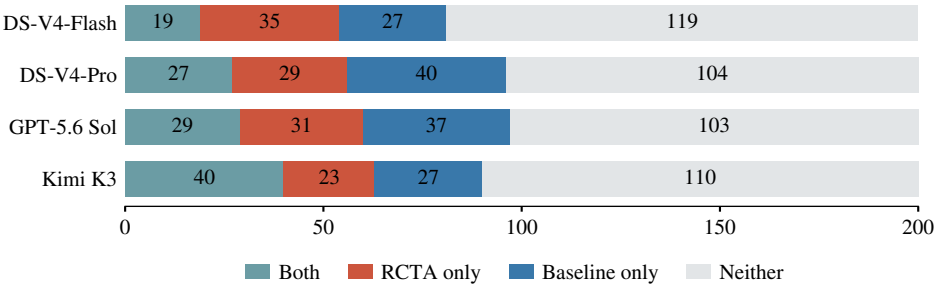


Figure 8: Exact-root correctness on LongRCA-Mini ($n = 200$ per backbone). Both: RCTA and at least one of five baselines are correct; RCTA only: RCTA is correct and all five baselines are wrong; Baseline only: RCTA is wrong and at least one baseline is correct; Neither: all six methods are wrong. Segment labels are trajectory counts.

Across all four backbones, 26 trajectories have at least one correct RCTA prediction and no correct prediction from any of the 20 baseline method–backbone pairs. Conversely, 43 have a correct baseline prediction and no correct RCTA prediction; 75 have successes from both method sets, and 56 have none. These counts summarize differently sized sets of observed predictions, not an ensemble comparison. Among the 26 RCTA-only trajectories, 18 have non-handoff reference roots. Inspecting all recorded role names further identifies 15 trajectories without any explicit handoff marker: eight from TravelPlanner, four from VitaBench, and three from WebArena Verified. The absence of the explicit marker does not establish the absence of implicit coordination or identify which component produced the successful attribution.

Illustrative roots with and without explicit handoffs. Two selected 162-step trajectories illustrate the observed coverage. In `swe_bench_pro__071`, the reference root is a handoff at step 27: the instruction requests a new `repoquery` parser but omits connecting it to `parseInstalledPackages` for the six-field output. In `travelplanner__458`, the reference root is step 84, where the accommodation agent proposes using estimated industry-average prices after its tool results fail to provide suitable listings for the required cities. This trajectory contains no explicit handoff marker. RCTA exactly matches the reference step on both cases with all four backbones, whereas none of the 20 baseline method–backbone results does. The cases illustrate distinct root structures; they do not estimate the prevalence of task-specific failure types or constitute a controlled comparison of components.

C Supplementary comparison with Who&When

We evaluate all six methods on the complete Who&When dataset (Zhang et al., 2025) using DS-V4-Flash. The dataset contains 184 trajectories: 126 from algorithm-generated systems and 58 from hand-crafted systems. We compare these results with the full LongRCA evaluation, weighting each trajectory equally within its dataset.

The five baselines, as shown in Figure 1, obtain 30.4–36.4% Step Acc on Who&When and 2.8–13.2% on LongRCA. RCTA obtains 41.8% and 24.1%, respectively. It has the highest observed

accuracy on all three metrics on LongRCA. On Who&When, it leads in Agent Acc and Step Acc, while All-at-once leads in Step ± 5.

All six methods have lower accuracy on LongRCA for each of the three metrics in this evaluation. The median trajectory length is 10 steps in Who&When and 145 in LongRCA, so a five-step localization error has a different relative scale in the two datasets. We use Step Acc as the primary localization comparison and report Step ± 5 as a complementary measure. The datasets also differ in domains and workflows, so the cross-dataset accuracy gaps do not isolate the effect of trajectory length.

D RELEASED DATA SCHEMA

Normalization maps each retained execution to an ordered sequence of records covering multi-agent dialogue, planning, tool calls, observations, browser or terminal interactions, and evaluator feedback. Records preserve original content and roles, stable 0-based indices, and available tool and verifier metadata. Temporal order is preserved without synthesizing intermediate actions or subdividing source events into artificial steps. Annotators, reference labels, and evaluated methods share these indices within each trajectory; event granularity can still differ across sources.

Source evaluator outcomes guide failure selection and human annotation. Evaluated methods receive only `question_ID` and `history`, which includes the task instruction. Reference annotations are withheld during prediction, and only the responsible-role and root-step fields are scored. This field boundary does not remove evaluator or verifier feedback already recorded within the history.

Table 14: Released LongRCA Bench data schema. Evaluated methods receive `question_ID` and `history`, which includes the task instruction. The three reference annotation fields are withheld during prediction.

Field	Meaning
`question_ID`	Source-prefixed trajectory identifier
`history`	Ordered trajectory records containing `step`, `name`, `role`, and `content`; includes the task instruction
`mistake_agent`	Reference responsible role, scored independently of the root-cause step
`mistake_step`	Reference 0-based root-cause step, scored independently of the responsible role
`mistake_reason`	Trajectory-grounded annotation rationale, withheld during prediction and not scored

E SOURCE EXECUTION PROVENANCE

Table 15 summarizes task domains and agent organizations; Table 16 records workflow implementations and trajectory-generator assignments. Fixed-role teams, group-chat coordination, and sequential workflows differ in their role sets, handoff patterns, and placement of plans, tool results, and verification messages.

Source representation is uneven: TravelPlanner contributes 685 trajectories (60.1%), while the other four sources together contribute 455 (39.9%). MiniMax-M2.5 generated 681 trajectories, Kimi-K2.5 generated 276, and Qwen3.5-Plus generated 183. Only TravelPlanner and VitaBench contain trajectories from all three generators. These counts describe trajectory generation rather than the inference backbones used for attribution.

Table 15: Task domains and agent organizations of the retained LongRCA Bench trajectories. Counts are reported by source.

Source	N	**Task domain**	**Agent organization**
SWE-bench Pro	128	Software repair	Diagnosis–execution–verification team
Terminal Bench 2	42	Terminal tasks	Diagnosis–execution–verification team
TravelPlanner	685	Travel planning	Magentic-One group chat
VitaBench	108	Service-oriented tool use	Action expert–computer terminal workflow
WebArena Verified	177	Web interaction	Planner–web-surfer–critic workflow
Total	**1,140**	**5 task domains**	

Table 16: Execution provenance and generator counts for LongRCA Bench. The SWE-bench Pro and Terminal-Bench 2 execution records specify their custom role workflow and generator, but no separate third-party scaffold package. Generator columns correspond to MiniMax-M2.5 (M2.5), Kimi-K2.5 (K2.5), and Qwen3.5-Plus (Q3.5+). A dash indicates that the source contains no trajectory from that generator.

Source	N	**Workflow**	**M2.5**	**K2.5**	**Q3.5+**
SWE-bench Pro	128	Custom `DiagnostAgent–ActionAgent–JudgeAgent` team	128	–	–
Terminal-Bench 2	42	Custom `DiagnostAgent–ActionAgent–JudgeAgent` team	42	–	–
TravelPlanner	685	AutoGen `MagenticOneGroupChat` with specialist roles	265	260	160
VitaBench	108	`Action_Expert–Computer_terminal` workflow	69	16	23
WebArena Verified	177	AutoGen `Planner–WebSurfer–Critic` controller	177	–	–
Total	**1,140**		**681**	**276**	**183**

F LOCALIZATION BY SOURCE AND LOG CHARACTERISTICS

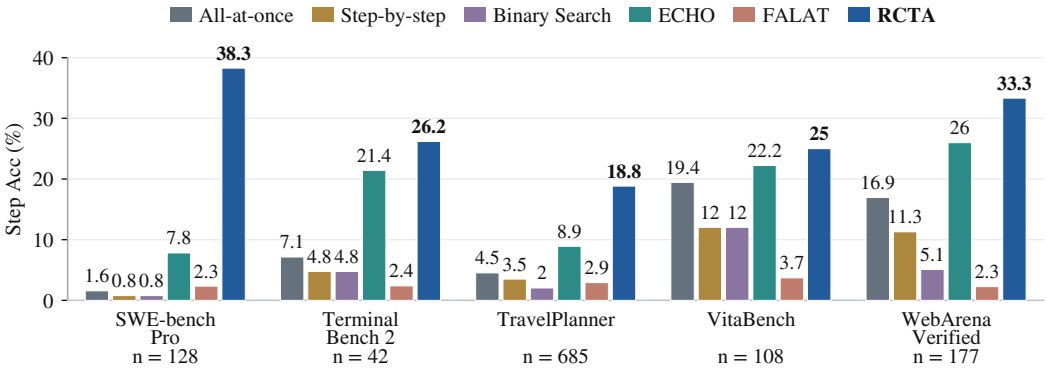


Figure 9: Step Acc (%) across five sources using DS-V4-Flash. All six methods use the same complete source subsets (n below source labels; total 1,140); invalid step outputs count as errors.

Table 17: Full-benchmark Step Acc (%) by source, underlying Figure 9. Source sample sizes are 128, 42, 685, 108, and 177, respectively. Every method uses the complete corresponding source subset; bold marks the highest observed accuracy in each source.

Method	SWE-bench Pro	Terminal Bench 2	TravelPlanner	VitaBench	WebArena Verified
All-at-once	1.6	7.1	4.5	19.4	16.9
Step-by-step	0.8	4.8	3.5	12.0	11.3
Binary search	0.8	4.8	2.0	12.0	5.1
ECHO	7.8	21.4	8.9	22.2	26.0
FALAT	2.3	2.4	2.9	3.7	2.3
RCTA	**38.3**	**26.2**	**18.8**	**25.0**	**33.3**

Table 18: Step Acc (%) across the five shared trajectory-length groups in Figure 5. The groups contain 290, 211, 212, 215, and 212 trajectories, respectively. Bold marks the highest observed accuracy in each group.

Method	10–63	64–129	130–172	173–220	221–728
All-at-once	21.4	6.2	2.8	0.9	1.9
Step-by-step	13.4	3.3	2.8	3.3	0.5
Binary search	7.2	3.8	0.5	2.8	1.4
ECHO	27.9	11.4	7.1	7.9	6.1
FALAT	4.1	3.8	0.0	3.3	2.4
RCTA	**34.8**	**18.5**	**21.2**	**18.6**	**23.6**

As shown in Tables 17 and 18, RCTA has the highest observed Step Acc in each source and each length group. These values underlie Figures 9 and 5. Both use the same full-benchmark predictions as Table 2. Each accuracy denominator includes all trajectories assigned to the corresponding source or length group.

Within the longest group, 170 of 212 trajectories (80.2%) have 221–400 steps. The RCTA–ECHO Step Acc difference in this group is 17.5 percentage points; it does not isolate performance on the extreme longest trajectories.

Shared length partition. The partition is constructed from one length per unique trajectory, with no prediction outcomes as inputs. Dynamic programming selects four cut points between distinct

observed lengths to minimize the sum of squared deviations of group sizes from $1{,}140/5$, while keeping equal lengths together. The selected boundaries are 63, 129, 172, and 220 steps. A concentration of 145 trajectories at exactly 63 steps limits how closely the groups can approach equal sizes. Five groups provide a descriptive resolution for the main figure.

Sensitivity to the number of groups. Repeating this length-only partition with four and six groups gives group sizes of 290/281/283/286 and 145/198/199/200/198/200, respectively. RCTA has the highest observed Step Acc in every group under all three choices, whereas local curve shapes vary. This sensitivity check supports the method comparison across these partitions while motivating caution when interpreting individual fluctuations. Source domains and workflows differ across the resulting groups.

F.1 WITHIN-SOURCE LOCALIZATION PATTERNS

Using the full-benchmark DS-V4-Flash predictions for all six methods, we examine localization within each source along two dimensions: root-to-end distance, $D = T-1-r^*$, and log text volume, measured by the total number of Unicode characters in history content, including whitespace. The former counts recorded activity after the reference root; the latter measures the amount of log text.

For each source and dimension, we choose the largest feasible number of groups up to three, with at least 30 trajectories per group and equal values kept together. Cut points minimize the sum of squared deviations from equal group sizes using only the grouping variable. All six methods share each group's trajectories, and invalid predictions count as errors. Terminal-Bench 2 has 42 trajectories and remains a single group. Figure 10 shows Step Acc; Table 19 reports both localization metrics. The accompanying analysis data include Wilson 95% intervals for group accuracies and Newcombe 95% intervals for each method's difference between the highest and lowest groups (Newcombe, 1998). These intervals are pointwise and conditional on the fixed groups.

Root-to-end distance shows heterogeneous patterns. On VitaBench, all six methods attain their highest Step Acc and Step ± 5 in the middle group; on TravelPlanner, differences between the highest and lowest groups have mixed signs across methods and metrics. Log text volume shows a more consistent contrast in these two sources: all six methods score lower in the highest-character group than in the lowest-character group on both metrics. Step Acc changes range from -8.8 to -0.9 pp on TravelPlanner and from -33.3 to -2.8 pp on VitaBench, with some intermediate-group reversals. Patterns on SWE-bench Pro and WebArena Verified vary by method. These source-specific associations do not establish a common monotonic trend or isolate a causal effect of subsequent activity or log text volume.

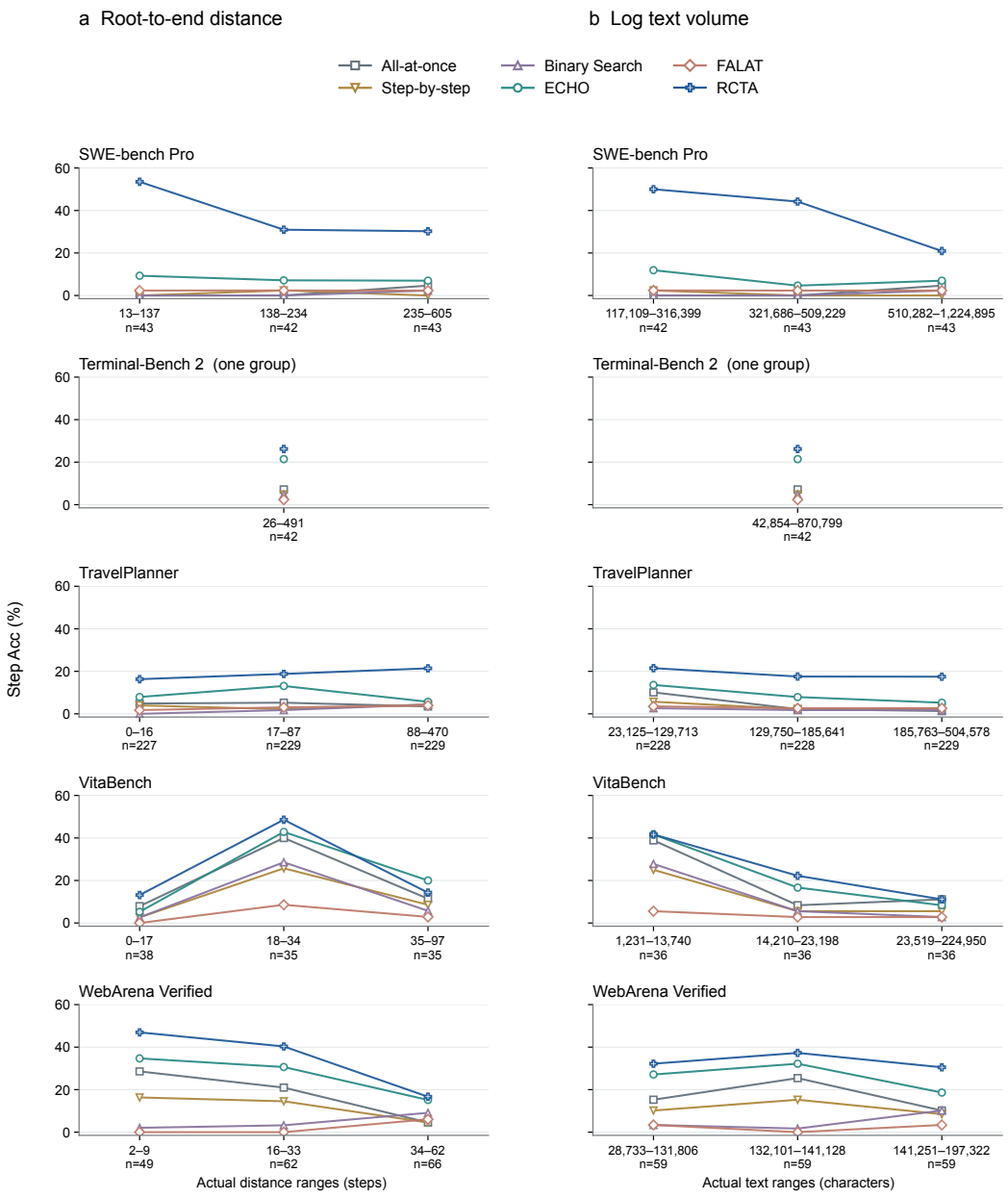


Figure 10: Within-source Step Acc with DS-V4-Flash by (a) root-to-end distance and (b) log text volume. All six methods share the trajectories in each group. Labels give actual ranges and sample sizes; unequal-width groups are equally spaced, and connecting lines are visual guides. Terminal-Bench 2 remains a single group under the minimum-size rule.

Table 19: Within-source localization with DS-V4-Flash. Each method cell reports Step Acc / Step ± 5 (%). All methods share the trajectories in each row. SWE, TB2, TP, Vita, and WA denote SWE-bench Pro, Terminal-Bench 2, TravelPlanner, VitaBench, and WebArena Verified, respectively.

Source	Group	Actual range	n	All-at-once	Step-by-step	Binary search	ECHO	FALAT	RCTA
Root-to-end distance (steps)									
SWE	1	13–137	43	0.0 / 7.0	0.0 / 7.0	0.0 / 4.7	9.3 / 18.6	2.3 / 2.3	53.5 / 62.8
	2	138–234	42	0.0 / 2.4	2.4 / 11.9	0.0 / 2.4	7.1 / 19.0	2.4 / 9.5	31.0 / 38.1
	3	235–605	43	4.7 / 4.7	0.0 / 11.6	2.3 / 4.7	7.0 / 16.3	2.3 / 20.9	30.2 / 39.5
TB2	1	26–491	42	7.1 / 21.4	4.8 / 16.7	4.8 / 23.8	21.4 / 38.1	2.4 / 16.7	26.2 / 45.2
TP	1	0–16	227	4.8 / 10.1	4.0 / 11.9	0.0 / 2.2	7.9 / 13.2	1.8 / 9.3	16.3 / 31.7
	2	17–87	229	5.2 / 15.3	2.2 / 9.2	1.7 / 10.0	13.1 / 23.1	3.1 / 10.5	18.8 / 27.9
	3	88–470	229	3.5 / 20.1	4.4 / 17.9	4.4 / 16.2	5.7 / 20.1	3.9 / 10.5	21.4 / 30.1
Vita	1	0–17	38	7.9 / 28.9	2.6 / 21.1	2.6 / 15.8	5.3 / 26.3	0.0 / 7.9	13.2 / 44.7
	2	18–34	35	40.0 / 57.1	25.7 / 42.9	28.6 / 54.3	42.9 / 60.0	8.6 / 48.6	48.6 / 60.0
	3	35–97	35	11.4 / 40.0	8.6 / 28.6	5.7 / 28.6	20.0 / 42.9	2.9 / 25.7	14.3 / 37.1
WA	1	2–9	49	28.6 / 57.1	16.3 / 42.9	2.0 / 8.2	34.7 / 46.9	0.0 / 12.2	46.9 / 69.4
	2	16–33	62	21.0 / 38.7	14.5 / 29.0	3.2 / 19.4	30.6 / 40.3	0.0 / 3.2	40.3 / 50.0
	3	34–62	66	4.5 / 16.7	4.5 / 18.2	9.1 / 31.8	15.2 / 30.3	6.1 / 22.7	16.7 / 39.4
Log text volume (characters)									
SWE	1	117,109–316,399	42	0.0 / 2.4	2.4 / 9.5	0.0 / 0.0	11.9 / 26.2	2.4 / 9.5	50.0 / 61.9
	2	321,686–509,229	43	0.0 / 7.0	0.0 / 11.6	0.0 / 7.0	4.7 / 9.3	2.3 / 4.7	44.2 / 53.5
	3	510,282–1,224,895	43	4.7 / 4.7	0.0 / 9.3	2.3 / 4.7	7.0 / 18.6	2.3 / 18.6	20.9 / 25.6
TB2	1	42,854–870,799	42	7.1 / 21.4	4.8 / 16.7	4.8 / 23.8	21.4 / 38.1	2.4 / 16.7	26.2 / 45.2
TP	1	23,125–129,713	228	10.1 / 22.8	5.7 / 18.0	2.6 / 14.5	13.6 / 27.6	3.5 / 17.1	21.5 / 38.2
	2	129,750–185,641	228	2.2 / 14.9	2.2 / 11.8	1.8 / 7.0	7.9 / 15.8	2.6 / 8.3	17.5 / 25.9
	3	185,763–504,578	229	1.3 / 7.9	2.6 / 9.2	1.7 / 7.0	5.2 / 13.1	2.6 / 4.8	17.5 / 25.8
Vita	1	1,231–13,740	36	38.9 / 72.2	25.0 / 55.6	27.8 / 55.6	41.7 / 66.7	5.6 / 52.8	41.7 / 80.6
	2	14,210–23,198	36	8.3 / 25.0	5.6 / 19.4	5.6 / 25.0	16.7 / 33.3	2.8 / 19.4	22.2 / 30.6
	3	23,519–224,950	36	11.1 / 27.8	5.6 / 16.7	2.8 / 16.7	8.3 / 27.8	2.8 / 8.3	11.1 / 30.6
WA	1	28,733–131,806	59	15.3 / 42.4	10.2 / 35.6	3.4 / 22.0	27.1 / 39.0	3.4 / 15.3	32.2 / 50.8
	2	132,101–141,128	59	25.4 / 40.7	15.3 / 28.8	1.7 / 20.3	32.2 / 39.0	0.0 / 8.5	37.3 / 50.8
	3	141,251–197,322	59	10.2 / 23.7	8.5 / 22.0	10.2 / 20.3	18.6 / 37.3	3.4 / 15.3	30.5 / 52.5

G ROOT-CAUSE DECISION RULES

The final-attribution stage applies the operational definition in Section 3 to the recorded trajectory. The rules below specify how it resolves common boundary cases and selects a single root-cause step.

- **Error already present in a handoff instruction.** If a later step implements an erroneous decision already stated in an earlier handoff instruction, the handoff step is selected as the root-cause step.
- **Error introduced after a handoff.** If a later step departs from the handoff instruction or introduces a decisive error not present in it, that later step is selected as the root-cause step.
- **Successfully repaired error.** An earlier error is excluded if it is corrected before the execution ends.
- **Error missed during later verification.** A later verifier may fail to detect an earlier error. This omission may be retained as unscored explanatory context, but it does not replace the step that introduced the error.
- **Contradictory task and evaluation requirements.** An input step may be selected only if the task instruction and the official evaluator or test requirement are explicitly incompatible. Both texts must be cited from the recorded trajectory.
- **Earliest supported introduction.** Among candidates supported by the original log, RCTA selects the earliest recorded step that introduced a decisive error relevant to the failed task execution. The selected error must remain unrepaired when the execution ends.

These rules govern only the root-cause-step output. RCTA predicts and scores the responsible role in a separate output field.

H RCTA PROMPTS AND IMPLEMENTATION DETAILS

At the implementation level, RCTA uses three prompt stages: local summarization, trajectory-outline aggregation, and final attribution. These implement, but are not identical to, the two scientific stages and four components in Figure 4. A *segment* is a consecutive block of original steps; a *phase* groups adjacent segment summaries serving the same subgoal without repartitioning recorded steps. A *handoff instruction* is a logged `X (-> Y)` message.

For segmentation, a *verifier signal* is a rule-extracted `PASS` or `FAIL` marker from explicit language in a verifier or terminal record. The signal serves only as an optional boundary cue, not as a semantic judgment, benchmark label, or the source evaluator's final outcome.

RCTA applies the local-summary prompt once per segment, followed by one trajectory-outline call and one final-attribution call. A trajectory with m segments therefore requires $m + 2$ initial LLM calls before any validation-triggered retry. All 1,140 full-benchmark RCTA predictions use the same three system-prompt templates for local summarization, trajectory-outline aggregation, and final attribution, provided with the implementation and the code for constructing user messages. The prompts contain task rules and output-schema guidance, but no LongRCA Bench instances or input–output demonstrations. User messages are constructed from the released identifier and trajectory history. The reference role, root step, and rationale fields are not provided during prediction.

The outline-stage validator in Table 20 checks referenced step identifiers. If the phase-construction response cannot be parsed, the implementation assigns one phase to each local summary.

The final-attribution validator checks the role and root fields independently. It accepts only a role label recorded in the trajectory and a root step drawn from the same trajectory. An invalid field does not overwrite a valid prediction in the other field. External subjects such as `ENVIRONMENT` are not benchmark labels and cannot be scored outputs. The marker `system_evaluation` may appear only as an unscored terminal node in the explanatory failure chain; it cannot be the predicted root step. Every supporting step must also belong to the recorded trajectory.

Segmentation uses rule-based implementation settings, not learned parameters. When adding the next step would exceed 32,000 characters, the implementation starts a new segment when possible;

Table 20: Prompt requirements and programmatic membership checks for the three RCTA prompt templates. The local-summary template is applied once per segment; the final role and root fields are validated independently.

Stage	Input	Required output	Validation
Local	Segment, task, and overlap	Purpose and candidate steps with evidence IDs	All IDs occur in the segment
Outline	Segment summaries and verifier-signal summary	Contiguous subgoal phases with step IDs	All IDs occur in the trajectory
Attribution	Outline, summaries, candidate text, and handoffs	Independent role and root fields, evidence IDs, and any cited handoff	Role and IDs occur in the trajectory; cited text is grounded

an individual overlong step may exceed this limit. Each segment contains at most 80 non-overlap steps. A boundary after a finish step, or before a role-changing handoff instruction or completion step, requires at least eight steps and 10,000 characters in the current segment. A boundary before a change in the most recent explicit verifier signal requires 10,000 characters but no eight-step minimum. After the 24,000-character target, an ordinary source-role change can form a soft boundary.

Except for the first, each segment receives up to five preceding steps from the previous segment as overlap context. The base candidate list is capped at 80 before preceding handoff instructions are added, so the final excerpt set can exceed 80. For an upstream attribution, the checker accepts a normalized substring match or sufficient word overlap between the cited text and the referenced handoff instruction. This check establishes textual provenance but does not independently verify that the cited instruction contains the same causal error.

Candidate ordering and role matching. Candidate hints are deduplicated by integer step ID, merging distinct role, source, and note fields. The fixed retrieval priority is possible manifestation (0), possible root (1), terminal context (2), and other candidates (3); a recovered-noise hint adds 3. If several tags occur, the first matching category in this order sets the base priority. Ties use ascending step ID. The first 80 candidates are retained before handoff augmentation; the expanded set is deduplicated without a second 80-step cap. Actual excerpts contain only IDs present in the trajectory. These rules prioritize evidence retrieval, not the final root-cause decision. They are implemented by `_add_candidate`, `_candidate_priority`, and `_collect_candidate_step_hints` in `analyze/global_reducer.py` in the released implementation.

Role classes are determined by recorded names in `preprocess/step_enricher.py`. Planner names are `DiagnostAgent`, `Task_Planner`, `Orchestrator`, and `manager_agent`; executor names are `ActionAgent`, `Action_Expert`, `WebSurfer`, `FileSurfer`, `Assistant`, `Coder`, `transport_agent`, `stay_agent`, and `food_attr_agent`; verifier names are `JudgeAgent`, `Verification_Expert`, and `verifier_agent`. Classification trims whitespace and extracts the sender from a handoff name. Names ending in `Expert` or `_Expert` map to verifier if they contain `verif` or `validat` (case-insensitive), and otherwise to executor. `Computer_terminal` and `ComputerTerminal` map to terminal, and `human` maps to human; unmatched names remain unknown. An `Orchestrator` status suffix retains the planner class.

For executor or verifier candidates, handoff augmentation requires the logged recipient to equal the candidate's raw agent name, with a case-sensitive comparison; matching the broad role class is insufficient. It selects the nearest strictly earlier matching handoff and uses no cross-recipient fallback. Other candidate roles use the nearest earlier handoff regardless of recipient. The relevant functions are `_delegate_target`, `_nearest_preceding_delegate`, and `_augment_with_preceding_delegates` in the same reducer module.

The implementation also contains role-name-triggered recall rules for a configured family of planner–executor logs. These rules can add visible executor errors, planner decisions, recovered-error context, and terminal context before candidate prioritization. None of the 1,140 LongRCA

Bench trajectories matched the configured role-name signature, so these rules did not affect the LongRCA results.

Ablation inference settings. Component comparisons use DS-V4-Flash with output budgets of 4,000 tokens for local summaries (reasoning off), 12,000 for aggregation, and 32,000 for attribution. One length-triggered retry doubles the budget, subject to a 65,536-token cap. Attribution requests temperature 0; aggregation and attribution use reasoning at high effort. All component conditions share these settings. Full-benchmark evaluation uses output budgets of 4,000, 6,000, and 8,000 tokens for local summarization, aggregation, and attribution, respectively.

Component-control implementation. The A3 control omits explicit backward-reasoning instructions while keeping the original candidate and handoff texts and step IDs. It removes linkage and role hints, replaces annotations for added handoff excerpts with neutral context labels, and omits the instruction-evidence check. The A4 control uses Full's initial attribution output after any length-triggered retry and omits validation-driven retry and output sanitization. Its output is checked only to measure validity.

Ablation statistical protocol. Component comparisons use source-macro aggregation over the five sources and four pre-specified exact McNemar tests on Step Acc against Full RCTA, with Holm correction (family α=.05). Source-stratified paired bootstrap CIs (10,000 resamples; seed 20260820) are pointwise and quantify trajectory resampling conditional on the observed outputs. At the observed discordance rates ($\approx$0.15), a post-hoc sensitivity calculation estimates 80% power for effects of roughly 9–10 percentage points in the four-test design. Accuracy uses a fixed denominator of 200; abstained, missing, or unparseable predictions count as errors.

Inference-time repeatability. Across three pipeline realizations on a 30-trajectory panel, Full RCTA obtains 26.7/30.0/43.3% Step Acc, compared with 26.7/16.7/30.0% without A1 and 23.3/20.0/33.3% without A2. The corresponding Full-minus-ablation differences are 0.0/13.3/13.3 and 3.3/10.0/10.0 percentage points. Pairwise agreement between Full root-step predictions is 50.0–56.7%.

Candidate-retention diagnostic. The base-candidate set and the handoff-augmented excerpt set are evaluated separately. Relative to Full, the equal-size uniform control changes base-candidate recall by -31.5 pp (pointwise 95% CI $[-39, -24]$), all-excerpt recall by -25.5 pp ($[-33, -18]$), final-evidence recall by -13.0 pp ($[-19, -7]$), and Step Acc by -6.0 pp ($[-11.0, -1.0]$). The three recall contrasts are exploratory and are not members of the four-comparison Step Acc test family.

Handoff-root diagnostic. On the 41 handoff-root cases within the 80-trajectory cohort (Appendix L.3), removing A2 or A3 lowers Step Acc by 26.8 or 19.5 pp, respectively; the unadjusted exact McNemar p-values are .0034 and .077. On the 39 non-handoff-root cases, both Step Acc differences are zero. Among non-abstaining, parseable in-range handoff-root predictions, late-prediction rates are 40.0%/60.0%/51.2% and mean signed errors (prediction minus reference) are +24.0/+32.4/+32.3 steps for Full/without A2/without A3. These subgroup diagnostics are exploratory.

I RCTA INFERENCE PSEUDOCODE

The pseudocode exposes the correspondence between the operational inference sequence and A1–A4 while retaining the implementation fallbacks described in Section H. Indices c and h denote steps s_c (a base candidate) and s_h (its retrieved handoff instruction). Handoff retrieval (A2) only augments the evidence package. The instruction–execution comparison and independent role/root prediction occur in COMPAREANDATTRIBUTE (A3). After the single validation-triggered retry, valid role and root fields are preserved independently, unsupported fields are removed, the output is flagged for review, and the system abstains only when an essential field remains invalid.

Algorithm 1 RCTA inference for a failed trajectory.

Require: Failed trajectory $\tau = (s_0, \ldots, s_{T-1})$ with a normalized, step-indexed history.
Ensure: Responsible role $\hat{\rho}$, root step $\hat{r}$, evidence E, or abstention.

Stage I: Failure-relevant candidate recall (A1)

$\mathcal{S} \leftarrow \textsc{EnrichAndSegment}(\tau)$
$L \leftarrow [\textsc{LocalSummary}(S_j) \mid S_j \in \mathcal{S}]$
$P \leftarrow \textsc{OrganizePhases}(L)$
if P is not parseable **then**
    $P \leftarrow \textsc{SingletonPhases}(L)$
end if
$B \leftarrow \textsc{RetrieveSourceSteps}(L, P)$
if $\textsc{PlannerExecutorSignature}(\tau)$ matches **then**
    $B \leftarrow B \cup \textsc{RuleBasedCandidates}(\tau)$
end if
$B \leftarrow \textsc{TopPrioritized}(B, 80)$
$C \leftarrow \textsc{CandidatePackage}(B, L, P, \tau)$

Stage II: Handoff-aware backward attribution (A2–A3)

for all $c \in B$ **do**
    if s_c is emitted by an executor or verifier **then**
        $h \leftarrow \textsc{RoleMatchedHandoff}(c, \tau)$
    else
        $h \leftarrow \textsc{NearestEarlierHandoff}(c, \tau)$
    end if
    $C \leftarrow \textsc{AttachIfPresent}(C, h)$
end for
$(\hat{\rho}, \hat{r}, E) \leftarrow \textsc{CompareAndAttribute}(P, C)$

Reliability layer: Validation (A4)

$v \leftarrow \textsc{ValidateFields}(\hat{\rho}, \hat{r}, E, \tau)$
if v reports an invalid field **then**
    $(\hat{\rho}, \hat{r}, E) \leftarrow \textsc{RetryWithFeedback}(P, C, v)$
    $v \leftarrow \textsc{ValidateFields}(\hat{\rho}, \hat{r}, E, \tau)$
end if
return $\textsc{SanitizeOrAbstain}(\hat{\rho}, \hat{r}, E, v)$

J EXTENDED RELATED WORK AND COMPARISON NOTES

Broader failure-characterization resources describe errors at levels that complement agent and step attribution. MAST organizes recurring failure modes, whereas TRAIL attaches error types and impact to execution spans (Cemri et al., 2025; Deshpande et al., 2025). MP-Bench accommodates multiple plausible failure-inducing steps (In et al., 2026). Among the resources in Table 1, Who&When Pro scales data collection through controlled error injection, while TraceElephant emphasizes complete execution observability (Liu et al., 2026; Chen et al., 2026).

RAFFLES uses iterative judge–evaluator feedback, whereas AgentRx grounds diagnosis in checked constraint violations (Zhu et al., 2026a; Barke et al., 2026). CDC combines Shapley-based agent attribution with temporal causal discovery for critical-step prediction (Ma et al., 2025). Diagnostic support can also extend beyond the given trajectory. FAMAS re-executes failed tasks and ranks actions using spectrum statistics across runs, while CORRECT retrieves error schemata distilled from previously annotated failures (Ge et al., 2025; Yu et al., 2025).

In Table 1, *Error Source* distinguishes failures arising during ordinary task execution (Natural) from failures constructed through deliberate error injection (Synthetic). Controlled task conditions, such as HORIZON's horizon settings, are distinct from deliberate error injection. *Diagnostic Targets* identifies the annotated object: Agent denotes the responsible agent or workflow role, Component an execution component, and Type a failure-mode or error-cause category. Step and Steps retain the localization targets summarized in the table's superscript notes; their annotation rules remain benchmark-specific.

Total (Reviewed) reports the number of released or analyzed trajectories, followed by the human-reviewed count in parentheses. Equal counts indicate review of all counted trajectories, but do not imply identical review or annotation protocols across resources. *Avg. length (steps)* reports each resource's overall mean using its own step-counting convention; – denotes an unreported mean. For Failure as a Process, both trajectory counts and average length refer to failed runs.

K BENCHMARK CONSTRUCTION AND SUPPLEMENTARY DETAILS

K.1 HUMAN ANNOTATION AND QUALITY CONTROL

The same annotator pool participated in calibration and full-scale annotation. Calibration refined how they navigated long histories, traced candidate errors, and documented evidence; the root-cause definition and decision rules remained unchanged. In both stages, annotators independently reviewed task instructions, complete step-indexed histories, and source evaluator outcomes, then submitted a responsible role, root step, and evidence-grounded rationale.

DeepSeek V4 Flash Preview generated factual activity summaries to support navigation. The prompt requested descriptions of recorded activities without root-cause judgments. Summaries remained separate from annotation fields; reference labels and rationales required support from task requirements and original trajectory records.

Full-scale submissions with verifiable role, step, and rationale fields number 1,343 over 1,041 trajectories: 747 have one annotation, 286 have two, and eight have three. The multiply annotated subset therefore contains 294 trajectories and $286 + 8\binom{3}{2} = 310$ unordered pairs, each from different annotators. Pairwise agreement is computed before label resolution, with role names normalized as in Appendix L.1. Exact role, step, and joint matches number 205, 125, and 121 (66.1%, 40.3%, and 39.0%). Calibration is reported separately in Table 21; the two sets are not pooled. Typical annotation time was 30–40 minutes.

The full-scale step gap has median 5 and interquartile range 0–46. Only 31 of the 185 non-exact pairs (16.8%) differ by 1–5 steps; nearby localization therefore explains a minority of exact-step disagreements. The median gap is 0 for same-role pairs and 42 for different-role pairs. These descriptive results concern the existing multiply annotated samples.

A valid submission must select an integer step within the trajectory range and a role present in the recorded role set. Its rationale must identify trajectory evidence supporting the proposed causal interpretation. Submissions with invalid references, evident annotation defects, or insufficient evidence are excluded; selected steps are not automatically changed to repair submissions.

For trajectories with multiple valid annotations, labels are compared field by field, and exact matches are retained as consensus references. Two experts resolve disagreements using the original task, complete history, source evaluator outcome, and written rationales under the attribution criteria. Final quality control re-examines every retained trajectory, including calibration cases, and excludes unresolved attribution ambiguity, invalid labels, or inadequate evidential support. Each released reference retains separate role and root-step fields with a supporting rationale.

Sensitivity to observed root-step labels. We assess root-step sensitivity on the 294 multiply annotated full-scale trajectories with predictions from all six methods. We first score against the final reference roots, then repeat scoring over 10,000 draws (seed 20260918). Each draw uniformly selects one complete observed human annotation per trajectory and uses its root step for all six methods. Trajectories and predictions remain fixed, and scores weight trajectories equally.

On the 294-trajectory subset, RCTA strictly leads all baselines in 100.00% of draws for Step Acc and 99.68% for Step ± 5; its gap to the strongest baseline has a central 95% range of [3.1, 8.8] and [1.4, 8.5] pp, respectively. Full method rankings vary more than the leading step method (Table 22). These results describe root-step sensitivity within the available annotation overlap.

Automatic release checks. All 1,140 released records have an existing integer root index, a role in the recorded role set, a nonempty rationale, unique contiguous step indices, matching individual and merged records, and a verified mapping to source trajectory content. These checks establish

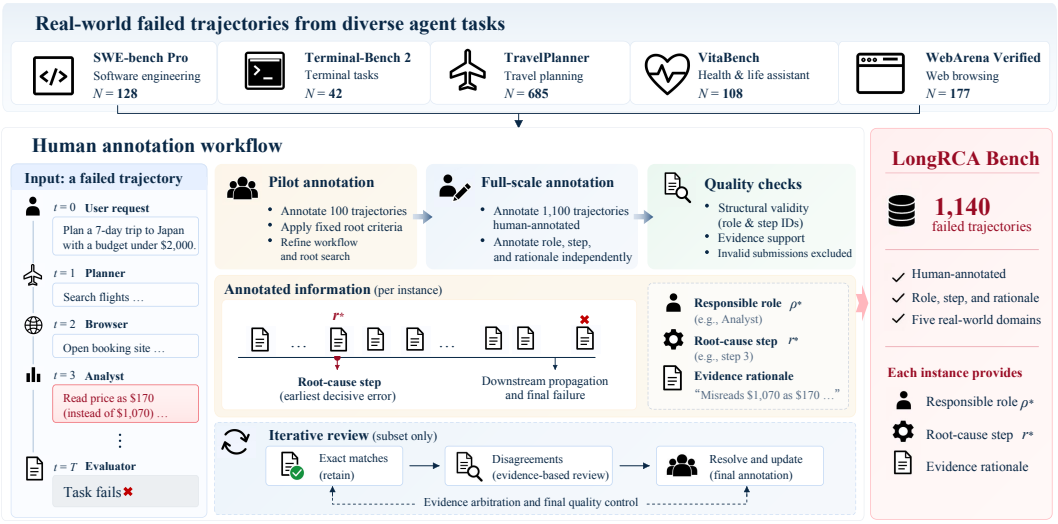


Figure 11: Construction and annotation workflow for LongRCA Bench. Source runs are filtered and normalized into step-indexed histories before annotation, label resolution, and final quality control.

Table 21: Pre-resolution disagreement among completed, structurally valid annotations. Each unordered pair of different annotators has equal weight. Role agreement uses the evaluation normalization. Step-gap percentages sum to 100 within each row; quartiles use linear interpolation. Calibration and full-scale annotations are summarized separately.

Role relation	Pairs	Role	Step ± 5	Median [Q1, Q3]	0	1–5	6–20	> 20
Full-scale: 294 trajectories								
All pairs	310	66.1	50.3	5 [0, 46]	40.3	10.0	11.6	38.1
Same role	205	100.0	67.3	0 [0, 15]	59.0	8.3	8.3	24.4
Different role	105	0.0	17.1	42 [13, 78]	3.8	13.3	18.1	64.8
Calibration: 100 trajectories								
All pairs	296	53.7	37.8	17.5 [1, 54.25]	24.7	13.2	15.5	46.6
Same role	159	100.0	59.7	1 [0, 29]	45.9	13.8	11.9	28.3
Different role	137	0.0	12.4	37 [15, 82]	0.0	12.4	19.7	67.9

Table 22: Root-step sensitivity to observed human labels on 294 full-scale trajectories with DS-V4-Flash. Accuracy cells show final-label score $\rightarrow$ resampled mean [2.5th, 97.5th percentiles] (%). Gap rows use RCTA minus the strongest baseline in each draw (pp). Rank-1 rates require a strict lead; ties do not count as wins. Intervals describe label sensitivity on fixed samples, not population confidence intervals.

Method / comparison	Step Acc	Step ± 5
All-at-once	7.8 $\rightarrow$ 8.3 [7.1, 9.5]	21.1 $\rightarrow$ 22.1 [19.7, 24.5]
Step-by-step	6.1 $\rightarrow$ 6.5 [5.4, 7.5]	15.6 $\rightarrow$ 17.3 [15.3, 19.4]
Binary search	3.7 $\rightarrow$ 3.9 [3.1, 4.8]	15.0 $\rightarrow$ 15.9 [13.9, 18.0]
ECHO	17.0 $\rightarrow$ 17.6 [15.6, 19.4]	27.9 $\rightarrow$ 31.1 [28.6, 33.7]
FALAT	1.0 $\rightarrow$ 2.2 [1.4, 3.1]	13.6 $\rightarrow$ 14.4 [12.2, 16.7]
RCTA	21.8 $\rightarrow$ 23.5 [21.1, 25.9]	35.4 $\rightarrow$ 36.2 [33.7, 38.8]
RCTA gap (pp)	4.8 $\rightarrow$ 5.9 [3.1, 8.8]	7.5 $\rightarrow$ 5.1 [1.4, 8.5]
RCTA strict rank 1	100.00%	99.68%
Full ordering retained	98.39%	72.15%

structural integrity and traceability, not causal correctness; neither agreement nor label sensitivity estimates final-label accuracy or independent expert reproducibility (Klie et al., 2024).

Responsible role and the recorded event. The role and root step are recorded separately, with a rationale grounded in the task and trajectory. A log event can relay another agent's action or completion report. For example, in `swe_bench_pro__047`, step 133 is logged under `Computer_terminal` but relays `ActionAgent`'s delegated completion report. The retained human reference assigns responsibility to `ActionAgent`, with step 133 as the root. Its rationale identifies the missing required configuration file despite the success report, and distinguishes later missed checks from that failure. This example shows why the event's outer sender does not mechanically determine the reference role.

K.2 Complete illustrative case

Figure 2 shows a pricing error in a 223-step TravelPlanner trajectory.[2] The user requests a three-day trip for one traveler from Plattsburgh to Fort Lauderdale within a $1,200 budget. The human reference assigns responsibility to `stay_agent` and identifies step 73 as the root-cause step.

The accommodation-search result at step 72 contains prices for multiple properties. In one subsequent output, `stay_agent` issues five candidate-recording calls, represented as steps 73–77 in the normalized history. Step 73 records Clean 1br at "$483 for 2 nights," and step 74 records Lefferts Garden 1 bedroom at "$317 for 2 nights." The remaining calls likewise interpret candidate prices as two-night totals. The reference step is the first candidate record in this output, while the displayed budget path follows Lefferts, the property ultimately selected. The calls document repeated misinterpretation within one output; the log does not establish a direct dependency from the step-73 candidate record to step 74.

Steps 78 and 79 confirm that both candidate notes were stored on the blackboard. At step 83, the agent compares the candidates and recommends Lefferts, calculating a $172 remainder using its $317 lodging total. The planner allocates this remainder to food and activities at step 86. The workflow verifier approves the budget and later rechecks the accommodation after transport changes, retaining the same two-night interpretation. During finalization, the manager corrects a separate arithmetic error in the food/activity allocation from $357 to $257, but leaves the lodging total at $317. The correction therefore repairs the allocation arithmetic while preserving the erroneous price interpretation.

The final itinerary at step 220 lists Lefferts for both nights without stating a lodging price. TravelPlanner defines `price` as a nightly rate, but the original accommodation-search response does not specify the unit. Applying this pricing rule to the recorded itinerary gives flight costs of $\$341 + \$255 = \$596$ and lodging costs of $\$317 \times 2 = \$634$. These items alone total $1,230, exceeding the budget before food or local transport. This budget check is part of retrospective diagnosis; the recorded source evaluator reports commonsense failures without executing the budget test. The case shows how an early misinterpretation can persist through verification and local correction, requiring the final plan to be read together with intermediate records and the applicable pricing rule.

K.3 Additional benchmark statistics

Median trajectory length ranges from 51 recorded steps for VitaBench to 257.5 for SWE-bench Pro; source medians for root-to-end distance range from 24 to 186 steps. Figure 3 reports the overall distribution summaries and threshold counts. Table 7 retains the total recorded step count, and Table 8 reports source-level means alongside the Mini comparison.

[2] Released as instance `travelplanner__021` in LongRCA Bench, corresponding to TravelPlanner test query 104, generated by Kimi-K2.5 with Magentic-One group-chat coordination (run 0001).

L SUPPLEMENTARY EVALUATION PROTOCOLS AND DIAGNOSTICS

L.1 SCORING AND EXPERIMENT PROTOCOLS

For the benchmark comparisons, role names are normalized for case, whitespace, and explicit handoff suffixes. Agent and step accuracies are computed from their respective output fields. Every expected trajectory remains in the denominators; invalid fields count as errors for their respective targets.

All component conditions use the same reference labels and scoring rules. Role matching normalizes case and whitespace while retaining handoff suffixes; abstentions count as errors.

Component coverage is the percentage of trajectories with non-abstaining outputs that pass the validation checks for their condition. These checks cover recorded role names, root-step validity, explanatory-chain structure, and cited handoff evidence. Appendix H specifies the A3 control's changes to these checks and the A4 control's use of validity checks only for scoring.

For the full-benchmark comparison, we also report **source-weighted valid-output root MAE**. Let $\mathcal{V}_b$ be the trajectories in source b with a valid, in-range numeric root-step prediction, N_b the total number of trajectories from that source, and $N = \sum_b N_b$. The metric is

$$\frac{1}{N} \sum_b N_b \left(\frac{1}{|\mathcal{V}_b|} \sum_{i \in \mathcal{V}_b} |\hat{r}_i - r_i^*| \right).$$

We average absolute root-step errors over valid predictions within each source, then weight these source means by the original source proportions. Missing or invalid step outputs count as errors for both root-step accuracy metrics but are excluded from the source means used for MAE. Exact matching remains the primary localization metric. A fixed step tolerance represents different fractions of trajectories whose lengths differ by an order of magnitude.

Fidelity evaluates the fixed Mini QIDs using the same DS-V4-Flash predictions as the full benchmark. Appendix H records prompt requirements and validation rules; Algorithm 1 gives the field-preserving fallback sequence.

L.2 FULL-BENCHMARK SUPPLEMENTARY METRICS

RCTA, as shown in Table 23, achieves 51.1% Agent Acc, 24.1% Step Acc, and 37.4% Step ± 5, with a source-weighted valid-output root MAE of 38.6 steps. Relative to ECHO, the strongest evaluated baseline on all four metrics, its accuracy gains computed from unrounded values are 23.6, 11.0, and 12.6 percentage points, respectively, and its MAE decreases by 11.8 steps. Despite these gains, the exact reference root is recovered in fewer than one quarter of the trajectories.

Role priors. The majority-role baseline always predicts the most frequent reference role within each source, using frequencies from the evaluated set (Table 24). RCTA matches this baseline on VitaBench and exceeds it on the other four sources.

L.3 COMPONENT ANALYSIS AND DIAGNOSTICS

Table 25 retains the complete metrics and pointwise intervals, which capture trajectory resampling and exclude inference-time variability. RCTA (all components) denotes the complete component configuration on the fixed sample (40 trajectories per source; sampling seed 20260821), evaluated under the ablation inference settings in Appendix H.

A1: Root-evidence recall. The candidate-information package retains more reference-root evidence than the equal-size uniform control (Figure 6(a)). Base-candidate recall is 19.5% for the control and 51.0% for Full; recall in the handoff-augmented excerpt set is 37.5% and 63.0%, respectively. The difference narrows at final evidence (25.5% versus 38.5%) and Step Acc (22.0% versus 28.0%). This contrast evaluates excerpt selection and the accompanying location and provenance cues as a single package.

Table 23: Full-benchmark results on all 1,140 trajectories. All methods use DS-V4-Flash. Agent Acc measures agreement with the reference responsible role. Root MAE is the source-weighted valid-output mean absolute step error. Higher values are better for the three accuracy metrics, whereas lower values are better for MAE.

Method	**Agent Acc↑**	**Step Acc↑**	**Step ±5↑**	**Root MAE↓**
All-at-once	26.2%	7.6%	19.9%	55.9
Step-by-step	22.2%	5.3%	16.9%	52.3
Binary search	23.0%	3.4%	13.3%	61.7
ECHO	27.5%	13.2%	24.7%	50.4
FALAT	19.0%	2.8%	12.5%	66.6
RCTA (ours)	**51.1%**	**24.1%**	**37.4%**	**38.6**

Table 24: Reference-role distributions and majority-role baselines on Full ($N = 1{,}140$). Role counts are listed in descending order. K is the number of distinct normalized roles in each history, reported as median [range] when variable. Majority and RCTA columns report Agent Acc (%); Overall weights trajectories equally. Source abbreviations follow Table 2.

Source	K	**Reference roles (counts)**	**Majority**	**RCTA**
SWE	5	`DiagnostAgent` (79), `ActionAgent` (41), `JudgeAgent` (4), `human` (3), `Computer_terminal` (1)	61.7	63.3
TB2	5	`DiagnostAgent` (20), `ActionAgent` (18), `JudgeAgent` (2), `Computer_terminal` (1), `human` (1)	47.6	57.1
TP	6 [3–7]	`stay_agent` (174), `manager_agent` (162), `Task_Planner` (103), `transport_agent` (84), `verifier_agent` (67), `food_attr_agent` (56), `Computer_terminal` (39)	25.4	41.2
Vita	2	`Action_Expert` (89), `Computer_terminal` (19)	82.4	82.4
WA	4	`Action_Expert` (97), `Task_Planner` (43), `Verification_Expert` (29), `Computer_terminal` (8)	54.8	60.5
Overall	–	–	40.3	51.1

Table 25: Component ablations on the fixed 200-trajectory sample. Rates are source-macro percentages. MAE uses non-abstaining outputs with parseable in-range steps (n=196/188/196/197/192); Cov. uses all trajectories as its denominator and requires non-abstaining outputs that pass the condition's validation checks (Appendix L.1). Δ is ablation Step Acc minus that of RCTA (all components). Intervals are pointwise; p_{Holm} controls the four-test Step Acc family.

Setting	Agent Acc↑	Step Acc↑	Step ±5 ↑	MAE↓	Cov.↑	Δ [95% CI]	p_{Holm}
RCTA (all components)	56.0	28.0	41.0	28.3	97.0	–	–
w/o candidate-information package (A1)	53.5	22.0	39.0	30.5	91.5	-6.0 [-11.0, -1.0]	.17
w/o handoff evidence (A2)	53.5	24.5	38.0	29.7	97.0	-3.5 [-9.0, $+2.0$]	.75
w/o backward attribution (A3)	51.0	25.0	41.0	31.4	97.5	-3.0 [-8.5, $+2.5$]	.75
w/o validation (A4)	54.0	26.5	39.5	29.2	89.0	-1.5 [-3.5, 0.0]	.75

A2–A3: Handoff evidence and backward attribution. The diagnostic includes all 80 trajectories in the component sample with an explicit logged handoff: 40 from SWE-bench Pro and 40 from Terminal-Bench 2. The reference root, independently of model predictions, defines 41 handoff-root cases (25 SWE-bench Pro, 16 Terminal-Bench 2) and 39 non-handoff-root cases (15 and 24, respectively). Both groups contain handoffs; the other 120 trajectories are outside this diagnostic.

Full RCTA and the A2 and A3 ablations obtain 51.2%, 24.4%, and 31.7% Step Acc on handoff roots, respectively, and 28.2% each on non-handoff roots (Figure 6(b)). Removing A2 omits the

added handoff evidence; removing A3 retains the original excerpts while removing the backward-attribution package and yields more late predictions on handoff roots. The observed accuracy differences are concentrated in the handoff-root group within the shared diagnosis–execution–verification workflow. Figure 6 summarizes these exploratory contrasts; Appendix L.1 specifies the scoring rules.

A4: Output validity. Validation changes the final prediction object on 18 trajectories and the parsed root on 9; exact-root correctness changes on 3 of 200, all favoring Full. Valid-output coverage increases from 89% to 97%.

Appendix H reports the statistical protocol and the three-realization repeatability panel.